MATH 346 Project

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0.1 Executive Summary

This research project analyzes how the Sustainable Farm at Saint Mary's College is serving the community within the College, especially the faculty, staff, and students, focusing on the satisfaction level of people and what they want out of the market. We collected data from the community of Saint Mary's College by building survey with the help of our Statistics class and Dr. Kristen Kuter. We organized the data into questions and categories to better answer the questions posed to us by the Sustainable Farm about the demographics, based on the role in the community and cultural background, and how people are satisfied with the Farm, and what hinders them from not attending the Market. As it operates now, the market meets a majority of the community's expectations, but there are some areas that need improvement. While satisfaction levels are generally high, variety and amount of produce could be improved slightly. Additionally, the time and location were inconvenient for a lot of respondents. There was not enough data to accurately compare some of the demographic data with the questions about people's experiences with the farm.

0.2 Introduction

The sustainable Farm at Saint Mary's College operates a free-will-donation market every Friday afternoon during the growing season, offering fresh, sustainably grown produce to the Saint Mary's College community, including staff, faculty, students, and others in the community. The farm's mission is not only to provide high-quality produce but also to contribute to social justice by ensuring that food is accessible to individuals from diverse backgrounds, regardless of their financial means. In line with this mission, the farm is interested in assessing the effectiveness of the market and understanding how well it meets the needs of various groups within the community.

In this report, we seek to analyze the farm's market through a survey conducted by our Statistics class, Math 346 taught by Dr.Kuter. The objective of the survey is to gather data on community members' experiences, satisfaction levels, and preferences regarding the market's offerings, depending on if they have been to the market or not. Specifically, our project aims to answer several key questions:

1. How well is the farm serving different demographic groups at the College, including staff, students, and faculty?
2. How well is the farm serving visitors from various income levels and cultural backgrounds?
3. How does satisfaction with the market vary across different community roles, income levels, or cultural/ethnic backgrounds?

4. What are the community's desires and expectations for the market's offerings, including the types of produce available, market time, and the donation model?

These questions are crucial for evaluating how well the market is fulfilling its mission of inclusivity and community support. Our goal from this data analysis is to provide them with ways that they can better fulfill their mission to help our community. Additionally, we aim to test hypotheses related to teh farm's ability to meet the diverse needs of the campus population, as well as to identify areas for improvement or potential changes in the market's operation.

The data for this analysis was collected through two primary methods: an in-person survey conducted at the market for people who attend, and there was also an electronic survey conducted online, distributed through the community of Saint Mary's College. Our focus in the report will be on the data collected from the community in general, whether or not they attended the market before. By analyzing the data, we hope to provide attainable recommendations that can guide the farm's efforts to improve its market and better serve the diverse needs of the Saint Mary's community.

0.3 Methodology

The survey design process for this project was carried out through the collaboration of all the students in the class, Dr.Kuter, and the Sustainable Farm. The first step in the survey design process was to identify important research questions and determine the types of data we needed to collect based on the questions posed by the Sustainable Farm. Based on the objectives outlined by the farm, we focused our questions in the survey to address the demographics of the community, including role, income level, and cultural background. We also addressed how the people experienced the market and their level of satisfaction with it. Finally, we also wanted to address the reasons for attending or not attending the market. In order to create the questions, we collaborated with Dr.Kuter to formulate the survey based on all the questions created by the class. When we were creating the survey, we also timed how long it took to make sure that the survey was not too lengthy to complete.

After creating the survey, we aimed to collect data from the whole Saint Mary's community, including staff, faculty, and students who had and had not attended the market to see how the market was making an impact on the community in general. To collect the data from the community, the three community groups in the class, made up of 3-4 students separated who the wanted to target with the survey. Our group focused on getting faculty to fill out the survey by having the administrative assistants for each department in the College to send out an email with the QR code to fill out the survey to all faculty in their department. This approach was chosen because we decided that faculty tended to pay more attention to emails sent by their administration. The other groups that focused on the community had emails sent to the whole community by our professor, Dr.Kuter, and posters were hung in all bathrooms to target the students on campus. By combining these approaches, we wanted to gain a wide variety of answers from different demographics.

There were several challenges arose throughout this process. First, we encountered problems with people understanding the survey, causing less people to fill it out. Because it was done over email and through posters, people that spoke a different language possibly had difficulty filling out the survey, so we did not get as wide of a demographic as we intended. While we did have this survey in Spanish, since there was no one to explain it as there were at the market survey, it was more confusing through the methods we used. Furthermore, because of the topic we had, we were concerned that people who had not attended the market would not be as interested in filling out the survey, making the data have a misrepresentation of people who have or have not attended the market.

After the data was collected and organized by our professor, Dr.Kuter, we separated the data into text-based answers, which were open ended, from the other data that was categorical, but used numbers to represent the different categories. We also combined certain demographic categories (race and ethnicity) to make the data more user friendly. Furthermore, we combined people that selected multiple ethnicity as multiethnic since there were not many people who selected multiple options. We also combined the student job status into one once, which allowed us to easily sort through the data. These data preparation steps made it easier to analyze the data and use EDA and hypothesis tests to understand the questions posed in our introduction and to gather meaningful insights into the effectiveness of the farm's free-will-donation market, with particular attention to the demographic groups at the College and the satisfaction level of these demographics.

0.4 Data Analysis and Results

After cleaning and preparing our data for analysis we had a sample of 100 responses to our survey. In our analysis we looked at a variety of questions from the survey in comparison to different demographic breakdowns of our sample. We will first do some exploratory data analysis to get a feel for our sample, before exploring the responses to the questions we analysed.

We first look at the Community Role and Race/Ethnicity categories. In terms of community role we have a fairly even spread across different roles. With 40 students, 36 faculty, and 24 staff responding. As we can see in Figure 1 the majority of respondents are white.

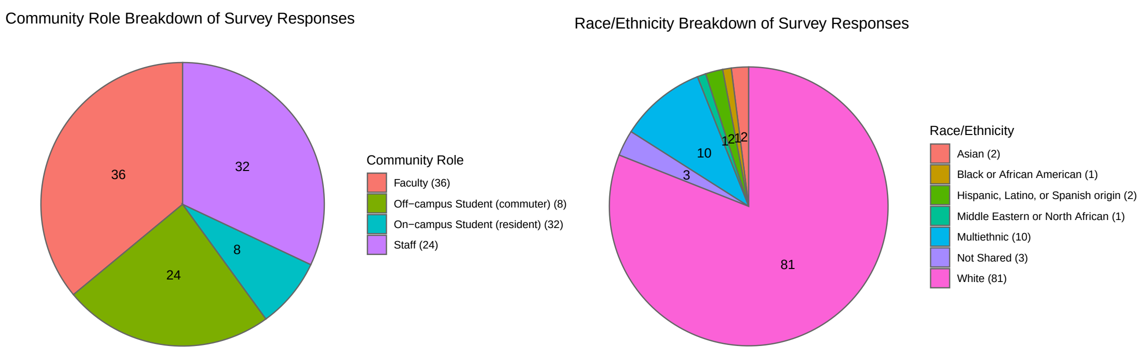

Figure 1: Community Role and Race/Ethnicity Breakdown

To investigate socioeconomic class we asked a couple of demographic questions. We asked faculty and staff about their income bracket, and we asked students about their job status and whether or not they have a meal plan. In terms of income, the faculty and staff responses were fairly evenly divided. The \$60,001 - \$90,000 bracket held the majority of responses with 20 out of 56 responses in that category. The majority of student respondent hold a job with 26 out of 39 respondents holding a job in some capacity. We also found that the vast majority of students have a meal plan with only 6 student respondents not having a meal plan. These statistics can be found in Figure 2.

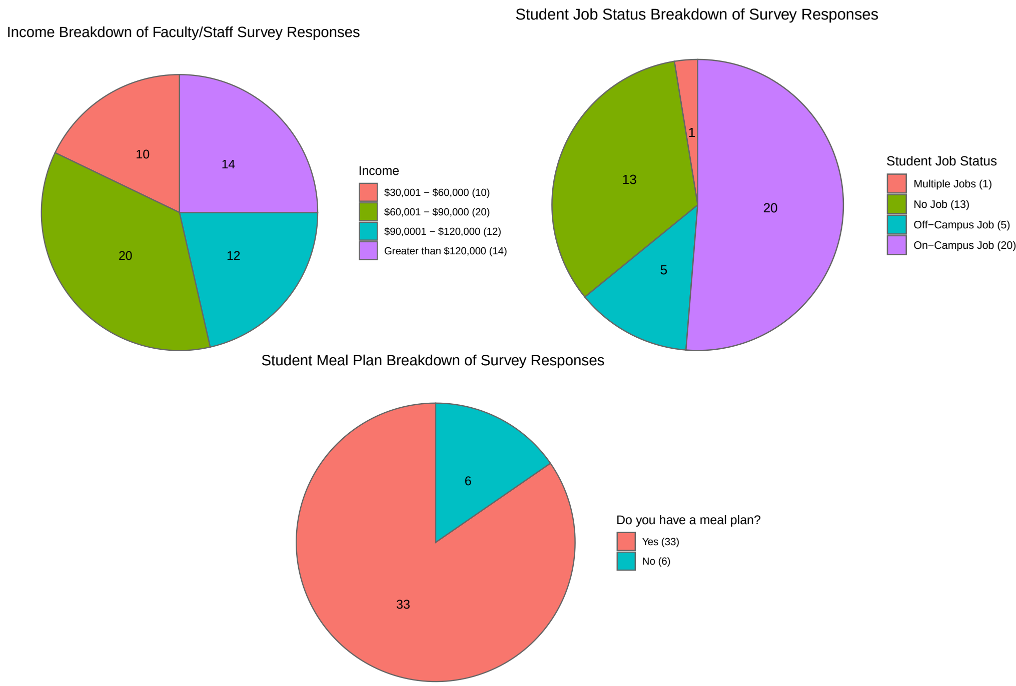

Figure 2: Socioeconomic Indicators Breakdown

0.4.1 Q10 - Methods of Hearing about Market

The next question from the survey we look at was Question 10, which asked about how people heard about the market. It seems that the majority of people have heard about it from emails and by word of mouth. 99 of the respondents answered this questions, 94 of which have heard about the Market through emails, while 62 respondents report hearing about the Market through word of mouth. These responses are plotted on a bar graph in Figure 3.

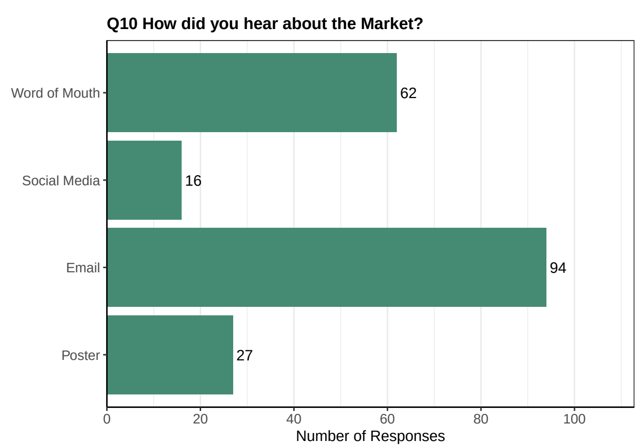

Figure 3: How People Have Heard of Market

We next separated the distribution by roles (staff, faculty, on-campus students, and off-campus students) to see any patterns in how each group heard about the market, shown in Figure 4.

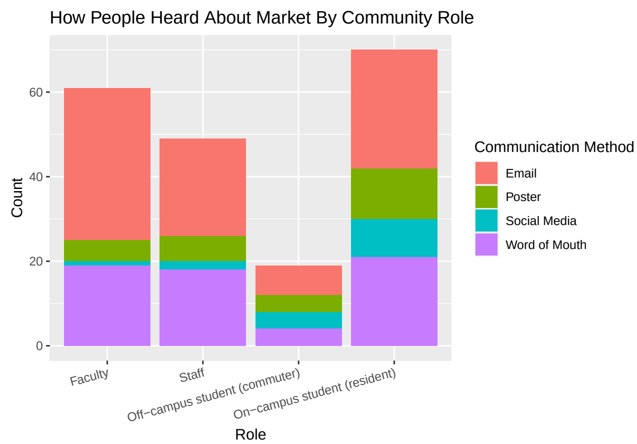

Figure 4: How People Heard About the Market Sorted by Role

Based on this bar graph showing the distribution count, it appears to show that word of mouth and email are the ways that most people in general hear about the market. However, this is not an exact showing of the variables since there is overlapping in the answers, and people could answer that they heard the market through multiple ways. However, the way that the data was separated to account for all options was to make each response its own independent person. Thus, the count is for all responses and does not represent the amount of people that answered each response since there is overlapping in the people that responded. Therefore, to determine how many people answered more than one choice, we looked at a table count shown in 4.3.1.

0.4.1.1 1.Table of All Combinations For Hearing about Market

Combination	Count
Email	31
Email + Social Media	1
Email + Social Media + Word of Mouth	3
Email + Word of Mouth	34
Poster	1
Poster + Email	2
Poster + Email + Social Media	2
Poster + Email + Social Media + Word of Mouth	9
Poster + Email + Word of Mouth	12
Poster + Social Media + Word of Mouth	1
Word of Mouth	3

Based on the table, the people that answered more than one answer usually answered email and word of mouth. There also are people that answered other options that are more than one. Yet, based on the table, we could determine the main way that people heard about the market was through email as shown by the 31 count. From this data, we performed a χ^2 test based on these two hypotheses.

$$H_0 : \text{Role and the method of Communication are independent from each other.}$$
$$H_A : \text{Role and the method of Communication are dependent on each other.}$$

However, it violates the independent assumption for χ^2 tests, so we could not fully conclude anything based on this result without caution. Our p-value used with a permutation test since there was not enough data for each category was 0.04654. Thus, while this p-value is significant, meaning that we could reject the null hypothesis that the role and method of communication are independent, we cannot say for certain that they are dependent on each other since this data set breaks an important rule in χ^2 tests, which is that the data must be independent. In order to make this test more reliable, we removed word of mouth as an option because this was the main contribution to having multiple responses, especially in regards to word of mouth and email. After removing this option from the data frame, we re-did the bar graph of the distribution of the method of communication across different roles, shown in Figure 5.

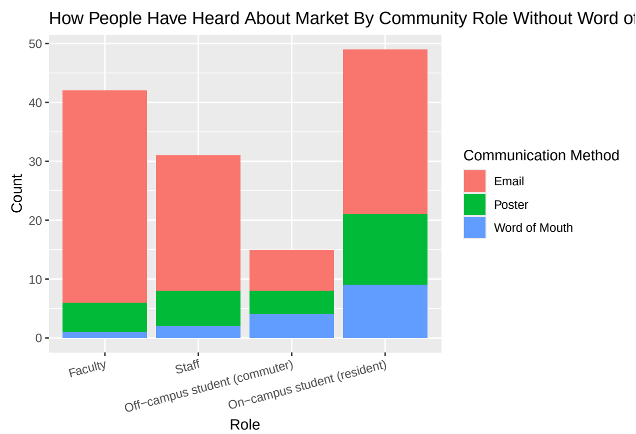

Figure 5: How People Heard About the Market Sorted by Role, Excluding by Word of Mouth

Based on this graph, the distribution is showing that email again is the main method that people have heard about the Sustainable Market. In addition to making this graph, we also ran the χ^2 test once again since there are less dependent and overlapping responses after word of mouth was removed. This change makes the test more reliable for the data since it does not violate the assumption as much; however, it still cannot be used as evidence without taking caution and finding out more information on the data. The p-value from this test was 0.0217, which is even smaller p-value, making it a significant value. This means that the null hypothesis can be rejected, but it is done with caution since this data was done with a χ^2 test that violates assumptions of this test. After looking at the proportions of the responses to roles, it does appear that email is used more by faculty and staff than students. However, the difference is not substantial, so we do not think that the χ^2 test is accurate since there are no noticeable patterns in relation to role. Furthermore, email is the highest used method for all roles in general.

```
##
##                                      email     poster socialMedia
##   Faculty                       0.85714286 0.11904762  0.02380952
##   Staff                         0.74193548 0.19354839  0.06451613
##   Off-campus student (commuter) 0.46666667 0.26666667  0.26666667
##   On-campus student (resident)  0.57142857 0.24489796  0.18367347
```

0.4.2 Q14 - Portion of Produce for Household

We then decided to look at the proportion of produce for a household that is being bought at the market. When looking at the portion of produce each person is using in their household from the market, we categorized it into 5 categories: less than 25%, between 25% and 50%, about 50%, between 50% and 75%, and more than 75%. We first created a bar graph of the general population's proportion of produce being bought in the market, shown in Figure 6. In this chart we can see that the vast majority of respondents are receiving less than 25% of their produce from the Market. There are 15 people who responded that they are recieving more than 25% of their produce from the Market.

```r
specialdata <- read.csv('CleanedDataRaceIncomeProp.csv')
proportion <- data.frame(specialdata$Q14)

proportion <- proportion[proportion != '',]
proportion <- data.frame(proportion)
```

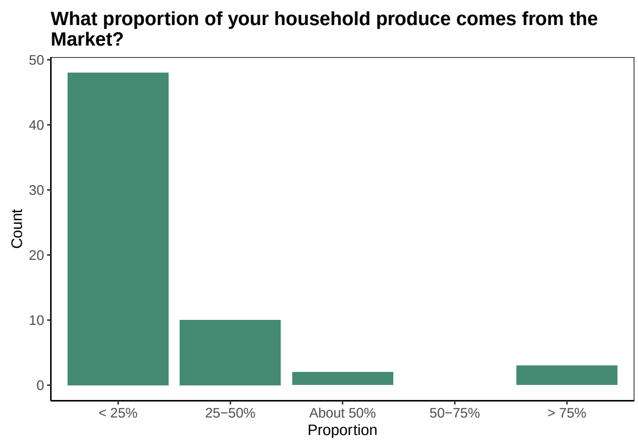

Figure 6: Proportion of Household Produce Coming From the Market for General Public

We next looked at the proportion of household produce in comparison to community role. To determine this relationship, we created a bar graph that calculated the counts for each of these categories in a specific role, shown in Figure 7.

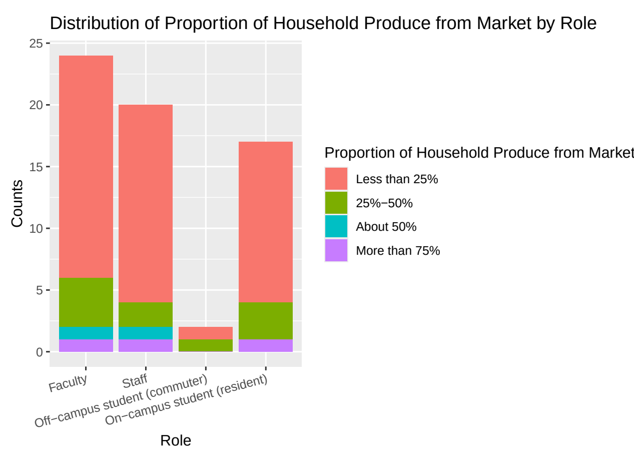

Figure 7: Proportion of Household Produce Coming From the Market Sorted by Role

Based on this graph, the majority of people had the market's supply less than 25% of all produce for their household, meaning that this was not the main means for people to get their produce. It did not vary based on role since most people in general found that it was not used for more than 25% of their produce. In addition finding the counts for each proportion category in each role category, we created a stacked bar graph to confirm these observations from the count bar graph, shown in Figure 8.

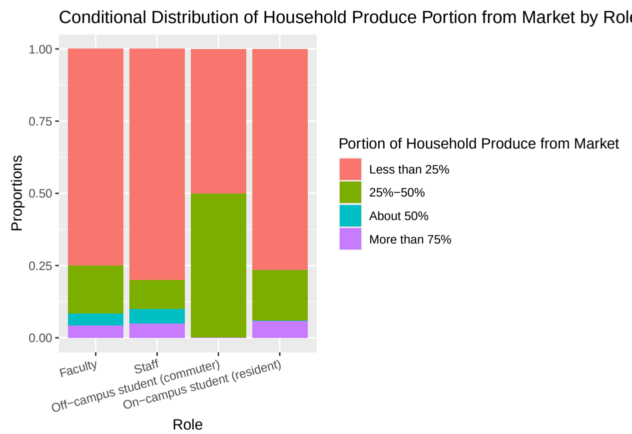

Figure 8: Proportion Breakdown of Proportion of Household Produce Coming From Market by Role

Based on these proportions, it appears that 0.75 of each role category agree that the portion of produce they collect is used for less than 25% of their total produce per their household. However, this value was similar across all categories, so it does not appear to show that there is a correlation between the role and portion of produce one gets from the market. These graphs show that in general, most people do not go to the market for all of their produce. To further validate this conclusion, we performed a chi-square test to determine if the null hypothesis can be rejected or not. The two hypotheses are:

H_0 : Role and the portion of produce people get from the market are independent from each other.
H_A : Role and the portion of produce people get from the market are dependent on each other.

The p-value from the chi-square test is 0.9552, meaning that we cannot reject the null hypothesis that states that the role a person has is independent from the portion of produce that person collects from the market. This value agrees with our earlier graphs that show that the role does not change the portion people usually get from the market.

We next compared the proportion of household produce to income. So we are only looking at responses from faculty and staff. In Figure @ref{fig:incomeProp} we see that the people who are receiving more than 75% of thier produce from the market are from the lower income brackets, but most people regardless of income bracket are receiving less than 25% of their produce from the market. We once again ran a χ^2 test to check to see if there is a significant relationship, and once again found no such relationship.

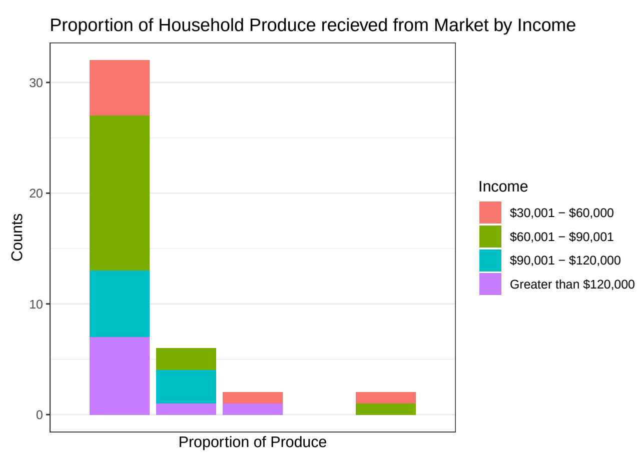

Figure 9: Proportion of Household Produce recieved from the Market by Income

```r
incomePropTab <- table(income, portion)
chisq.test(incomePropTab, simulate.p.value = TRUE, B = 10^5)
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  incomePropTab
## X-squared = 12.101, df = NA, p-value = 0.5894
```

0.4.3 Q16 - Satisfaction

Question 16 asked respondents who had attended the Market to rank their satisfaction on a scale of 1-5 in three different categories: quality, variety, and amount of produce. As we can see in Figure 10 overall satisfaction with these categories is generally high.

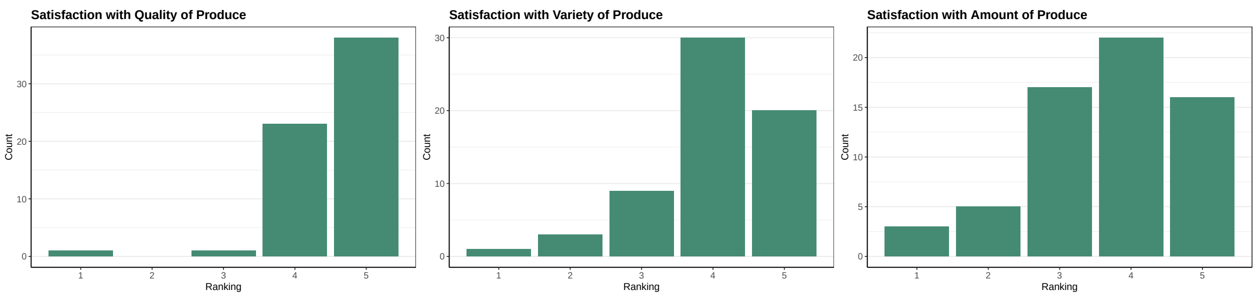

Figure 10: Satisfaction Level [1-5] For Categories of Product Attributes by General Public

We also analysed satisfaction in these categories in comparison to different demographic breakdowns in order to see if there is a certain demographic group that is more or less satisfied with the Market.

We first look at satisfaction broken down by role. There is no obvious patterns in the charts in Figure 11 indicating that satisfaction with the produce at the Market is not related to the role someone holds in the community. In order to confirm this suspicion we can run some χ^2 tests. In this context a χ^2 test looks at categorical data to see if two variables (here 'community role' and 'satisfaction with quality/variety/amount of produce') are independent or if they impact each other. In order to run a χ^2 test the sample must be of a certain size. This sample isn't quite big enough, but we can use permutations to still be able to perform the tests. The hypotheses we are testing in these tests are:

$$H_0 : \text{Community Role and Satisfaction with Produce are independent.}$$
$$H_A : \text{Community Role and Satisfaction with Produce are related.}$$

We run one test for each of the types of satisfaction we are looking at. After running the χ^2 tests we found that in fact Community Role and Satisfaction with produce are independent variables.

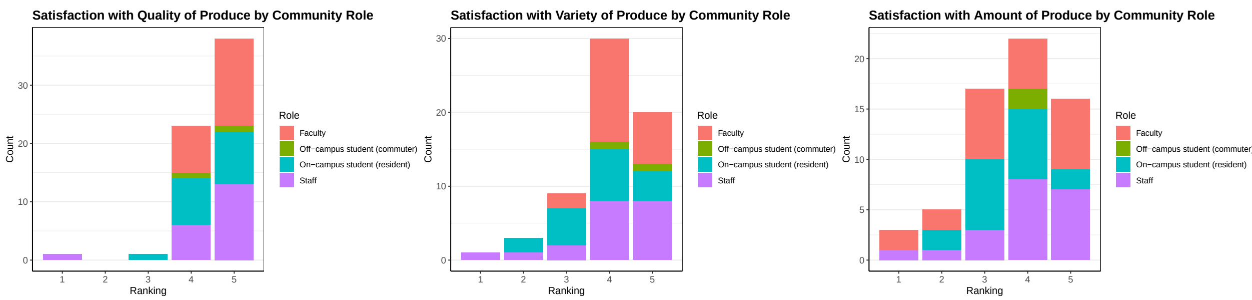

Figure 11: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Role

The next demographic breakdown we compared to satisfaction with produce was Race/Ethnicity. Once again the charts do not indicate any sort of pattern, shown in Figure 12. We ran χ^2 tests to confirm, and the results did confirm that race and satisfaction with produce are independent.

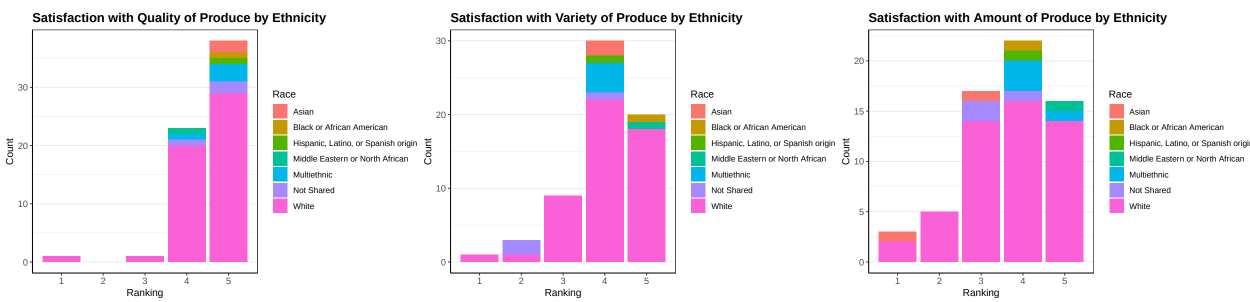

Figure 12: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Race and Ethnicity

The last set of demographic breakdown we compared produce satisfaction to were income (for faculty/staff) and job status (for students). We can first look at Figure 13 and see that the \$30,001 - \$60,000 bracket might be more satisfied on average. Sadly, in this case for satisfaction with quality and variety there was not enough responses to run a χ^2 test. The test for satisfaction with amount of produce showed that income and satisfaction with the amount of produce are independent.

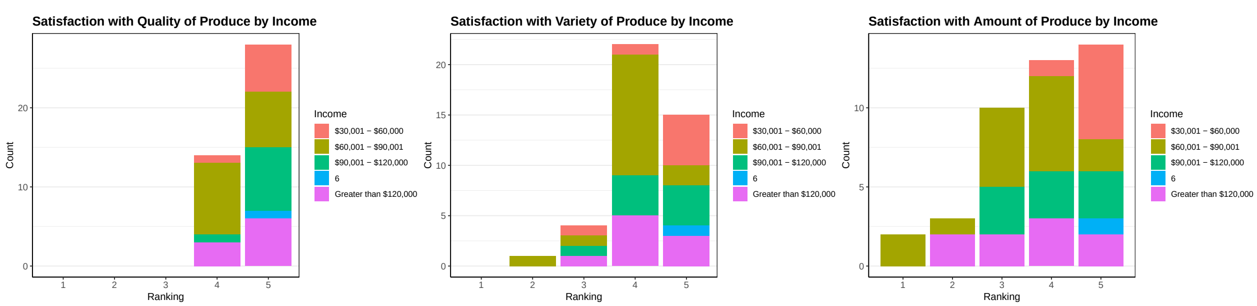

Figure 13: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Income

The last demographic breakdown we will compare produce satisfaction to is Student Job Status. In Figure 14 you will notice that there are only four bars in our bar charts. This is because no students rated their satisfaction a 1, so in order to perform χ^2 tests we dropped that column. From these plots we can see that students seem to be very satisfied with the quality of the produce.

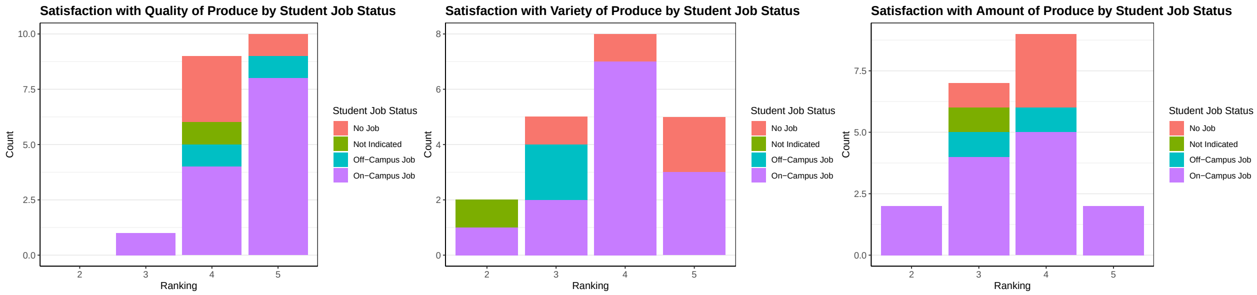

Figure 14: Satisfaction Level [1-5] For Categories of Product Attributes Sorted Job Status

We ran χ^2 tests to see if student job status and satisfaction with produce are independent. Even after dropping the 1 column there were not enough responses to analyse satisfaction with quality, but we were able to get results for variety and amount. Our hypotheses for these χ^2 tests were:

$$H_0 : \text{Student Job Status and Satisfaction with Variety/Amount of produce are independent.}$$
$$H_A : \text{Student Job Status and Satisfaction with Variety/Amount of produce are related.}$$

The test found that satisfaction with amount of produce and student job are independent, and that satisfaction with variety of produce and student job are related. The criteria for rejecting the null hypothesis and switching to the alternative hypothesis is a low p-value. The p-value gives the probability that the results we observed happened by chance. In the test on satisfaction with variety of produce and student job status we found a p-value of 0.03, which at an α-level of 0.05 is a significant result.

Since we found a significant result we did further analysis on the relationship between student job status and satisfaction with variety of produce. This analysis took the form of a proportion table. The proportion table showed that all of the students with off-campus jobs rated their satisfaction with the variety of produce a 3, which is likely why this test came back as significant. Even though this test is statistically significant, we do not believe that there is a meaningful take away since the result is likely due to one demographic happening to answer the same on this question.

```
##
##                          2          3          4          5
##   No Job         0.00000000 0.25000000 0.25000000 0.50000000
##   Not Indicated  1.00000000 0.00000000 0.00000000 0.00000000
##   Off-Campus Job 0.00000000 1.00000000 0.00000000 0.00000000
##   On-Campus Job  0.07692308 0.15384615 0.53846154 0.23076923
```

0.4.4 Q18 - People's Comfort Levels Paying

When analyzing people's comfort level not paying for produce, we analyzed it based on the person's role by doing a bar graph, shown in Figure 15.

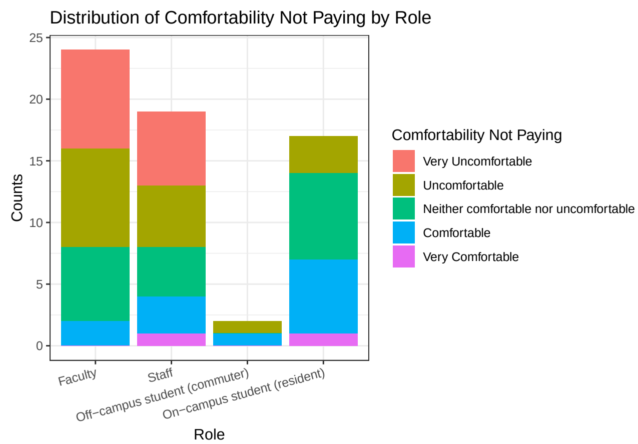

Figure 15: People's Comfort Levels Not Paying for Produce

Based on this graph, many people, especially faculty and staff, are not comfortable with not paying for produce. These two groups are very uncomfortable not paying, while the students are on a scale from uncomfortable to comfortable. Very few people are very comfortable paying, showing that age contributes to how comfortable people are not paying for things. College students are more comfortable with not paying than adults working, causing this divide in the data distribution. We also did a proportions test to further confirm our conclusions based on these results, shown below.

```
proportions(tableComfort, 1)
```

```
##
##                                         1          2          3          4
##   Faculty                      0.33333333 0.33333333 0.25000000 0.08333333
##   Staff                        0.31578947 0.26315789 0.21052632 0.15789474
##   Off-campus student (commuter) 0.00000000 0.50000000 0.00000000 0.50000000
##   On-campus student (resident)  0.00000000 0.17647059 0.41176471 0.35294118
##
##                                         5
##   Faculty                      0.00000000
##   Staff                        0.05263158
##   Off-campus student (commuter) 0.00000000
##   On-campus student (resident)  0.05882353
```

The proportions for this data also agrees with earlier statements that we made because the faculty and staff have the largest proportion of responses saying that they are very uncomfortable paying, and the students have the largest proportion saying that they are comfortable not paying.

0.4.5 Q22 - Social Justice Concerns

In order to see if the Market is meeting their social justice goals we asked survey respondents if they agree or disagree on a set of statements relating to inclusivity and the community impact of the Market. These nine statements were rated 1-5 by respondents who have attended the Market. In Figure 16, we can see the overall results of this question. In general people seem to agree that the Market is an inclusive and welcoming space that is an important resource for the community. People seem to think that the Market isn't as important a resource for them personally.

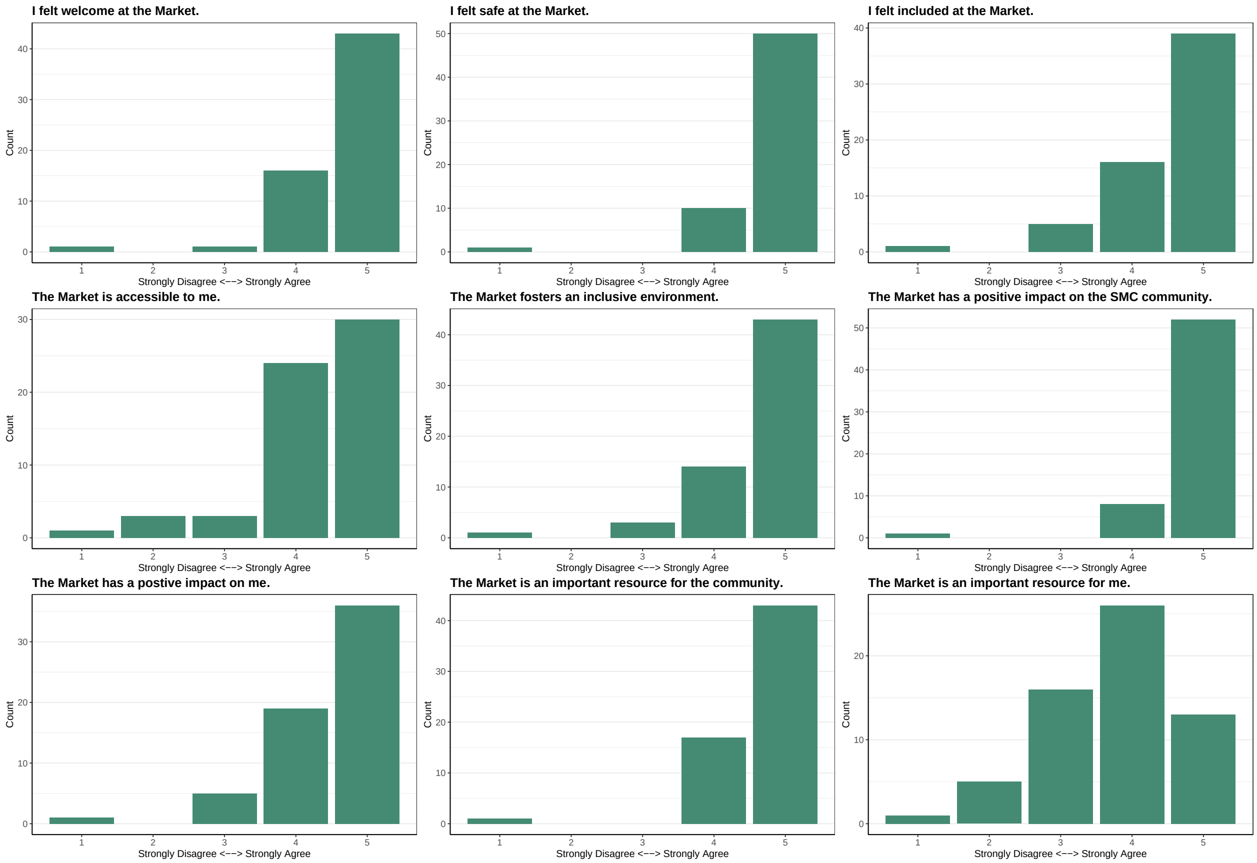

Figure 16: Satisfaction Level [1-5] on Opinion Statements of the Market's Goals by General Public

We next looked at how people are satisfied with different things that are involved in the Sustainable Market's goals in regards to different roles of people: staff, faculty, and students. First, we looked at people's satisfaction levels when it came to the accessibility of the farm by finding the median for each group of people and how their satisfaction level varied based on their role. We also created a bar plot that plotted the satisfaction level on the x-axis and the count for each different role on the y-axis stacked, shown in Figure 17. Furthermore, the medians for this data is under the variable AccMedian in the Appendix.

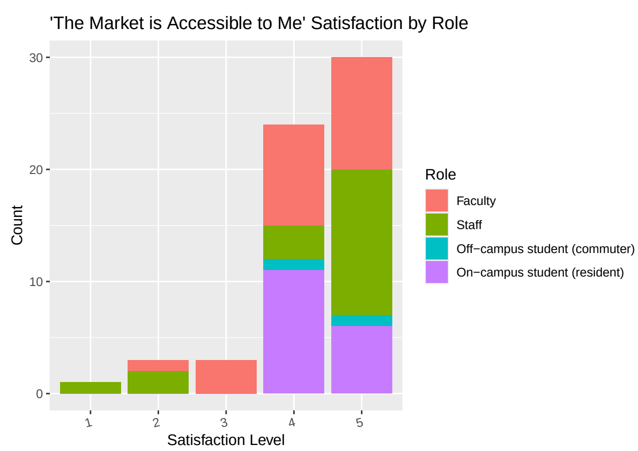

Figure 17: Satisfaction Level [1-5] For 'The Market is Accessible to Me' Statement Sorted by Roles

In general, the medians for each group are relatively the same, being fairly satisfied; however, the staff and faculty have lower satisfaction scores than the students. Their median scores are lower, showing that they are only satisfied, not very satisfied. This contribution could be due to them not wanting to give perfect scores, or there is something about the accessibility of the market that was missing. The bar graph of the distribution agrees with these previous statements of the faculty and staff not being as satisfied as the students when it comes to how accessible the market is. Because these groups of people have full-time jobs and are not always on campus, they could struggle with the time of the market or with its location because they are not always on campus. This can be further explained by finding the proportions of satisfaction in this category by roles, as shown below.

```
##
##                                       1          2          3          4
##   Faculty                    0.00000000 0.04347826 0.13043478 0.39130435
##   Staff                      0.05263158 0.10526316 0.00000000 0.15789474
##   Off-campus student (commuter) 0.00000000 0.00000000 0.00000000 0.50000000
##   On-campus student (resident)  0.00000000 0.00000000 0.00000000 0.64705882
##
##                                       5
##   Faculty                    0.43478261
##   Staff                      0.68421053
##   Off-campus student (commuter) 0.50000000
##   On-campus student (resident)  0.35294118
```

These proportions show that most of the satisfaction levels that disagree with the statement that the market is accessible to them are faculty and especially staff, whereas the students agree with this statement. This proportion agrees with all the other data that the faculty and staff do not agree with the market being easily accessible to them.

For the statement that the market is an important resource to the person personally, we again looked at the medians and bar plot of the satisfaction levels, grouped by the role of the person. The bar plot of the count for each satisfaction level sorted by role is in Figure 18. Furthermore, the medians for this data is under the variable ResourceMedian in the Appendix.

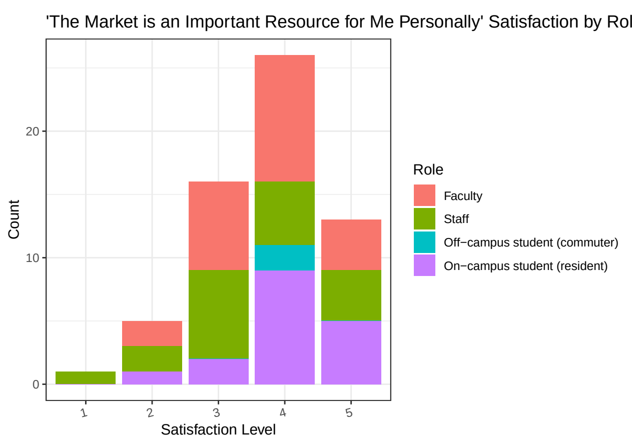

Figure 18: Satisfaction Level [1-5] For 'The Market is an Important Resource for Me Personally' Statement Sorted by Roles

In this case, the median for the satisfaction levels are lower than in the other questions, with a median score for most groups being either neutral or only agreeing with this statement, not strongly agreeing. This is especially prominent in the staff and faculty responses, with the staff having the lowest median satisfaction rating. We then looked at the proportions of each group to see how they compared to each other, shown below.

```
##
##                                        1          2          3          4
##   Faculty                     0.00000000 0.08695652 0.30434783 0.43478261
##   Staff                       0.05263158 0.10526316 0.36842105 0.26315789
##   Off-campus student (commuter) 0.00000000 0.00000000 0.00000000 1.00000000
##   On-campus student (resident)  0.00000000 0.05882353 0.11764706 0.52941176
##
##                                        5
##   Faculty                     0.17391304
##   Staff                       0.21052632
##   Off-campus student (commuter) 0.00000000
##   On-campus student (resident)  0.29411765
```

These proportion numbers agree with all previous statements made about this statement in relation to peoples' roles because most lower satisfaction levels are for staff. The satisfaction levels for on-campus students and faculty are similar, with mixed responses for agreeing with the statement. Finally, the off-campus students, who have the lowest response amount, agree with the statement. Thus, overall, the staff and faculty do not find the market as valuable a resource to them personally as the students do. This could be because they go to grocery stores regularly to get their produce, and students do not leave campus as often, so they are not buying much produce off campus.

The next demographic breakdown we compared the social justice measures to was Race/Ethnicity. We once again focused on the "The market is accessible to me" and "The market is an important resource for me" statements since they had a wider spread of responses. In Figure @ref{fig:raceXAccess} we can see that the minority race/ethnic groups were overall very much in agreeance that the Market is accessible to them, while responses from white respondents had a wider spread of responses though still quite positive. We did run a χ^2 test to see if there is a significant relationship between the variables, but the result was that there is not evidence to suggest a relationship.

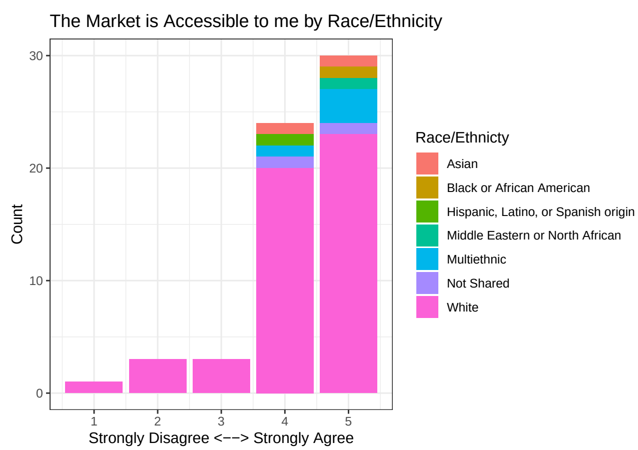

Figure 19: Agreement with the statement "The Market is Accessible to me" by Race/Ethnicity

0.4.6 Q26 - Why People Do Not Attend Market

We looked at why people could not attend the market by giving four options for people to choose from for why they do not attend: being unaware that there is a market, the time is bad, they are not interested, or the place is a bad location. To analyze this data in regards to different roles, we looked at a bar graph of all the options plotted across the different roles, shown in Figure 20. We ignored the factor of people selecting multiple options, and we chose to instead separate the responses into different variables, so one person could be counted multiple times if they had multiple responses, similar to how we did the method of hearing about the market analysis.

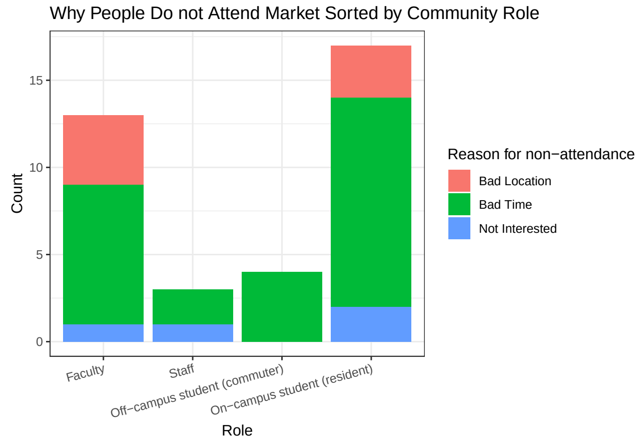

Figure 20: Why People do not Attend the Market

The bar graph shows that for everyone, no one has not attended because they were unaware that the market was taking place. Furthermore, from looking at the graph, it seems that most have not attended because they were uninterested or it has been at a bad time on Fridays. It seems also that the on-campus students have more reasons as to why they have not attended and struggle the most with the time because of classes or activities probably. We then looked at the proportions of the different responses across the roles to better understand the ratio of people responding with a certain answer.

```
proportions(table(df4_long$role, df4_long$Attendance), 1)
```

```
##
##                                  attendBadPlace attendBadTime
##   Faculty                            0.30769231    0.61538462
##   Staff                              0.00000000    0.66666667
##   Off-campus student (commuter)      0.00000000    1.00000000
##   On-campus student (resident)       0.17647059    0.70588235
##
##                                  attendNotInterested
##   Faculty                                 0.07692308
##   Staff                                   0.33333333
##   Off-campus student (commuter)           0.00000000
##   On-campus student (resident)            0.11764706
```

It appeared that the staff and off-campus students have not minded the location, and they struggle more with the time of the market since it is in the middle of the day during classes or people's work hours. The faculty and on-campus students struggle more with the location. Furthermore, the staff and on-campus students have been the least interested in attending the market because either it might be out of the way, or for students, they do not need the produce or have a place to store it.

0.4.7 What do people want to see from the Market?

Finally we looked at a few questions about what people wanted to see from the market. We asked respondents to indicate which categories of produce they are interested in. The categories we presented were vegetables, fruits, flowers, and herbs. All of our 100 respondents answered this question. We then plotted these results for the general population in a bar graph, shown in Figure 21. The most popular requests were vegetable by far with 93/100 respondents indicating interest. Fruit was also very popular with 83 people indicating interest, though herbs and flowers were also popular.

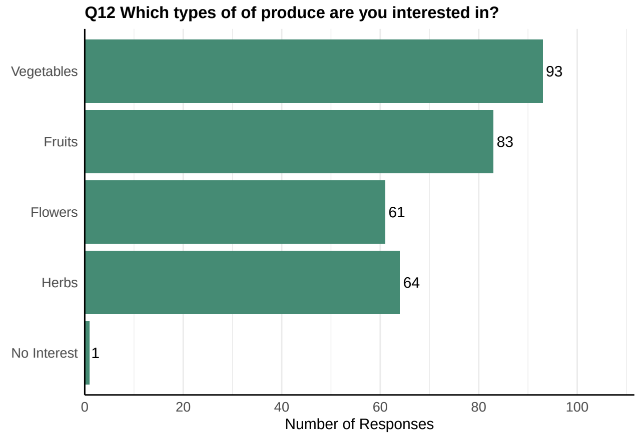

Figure 21: People's Interests in Different Type of Produce

There were also a couple of question on the survey that took text responses. These questions asked about what specific produce people are interested in and asked for recommendations for making the Market more accessible. In terms of produce the most prevalent responses were requests for fruit/berries, eggs, and potatoes. In terms of accessibility the most common responses were to hold the market in a more central location, to move the market off of the grass, and to have more or different times. All of the responses to these questions can be found in the Technical Appendix.

0.5 Discussion and Insights

A majority of all respondents are white, and a majority of students have jobs and meal plans. Community role (student, staff, or faculty) is mostly equally distributed. None of the faculty or staff said that they had a household income of less than $30,000, but the other four income brackets were mostly equal to each other. Because this survey was advertised to everyone in the Saint Mary's community, these demographics are a reflection of the community as a whole rather than specifically people who have been to or frequently go to the market.

In general, respondents seem satisfied with the market and believe it to be beneficial to the community. Satisfaction of produce variety, quality, and quantity is very high. Because there were not enough responses, in many cases, it was not possible to determine if race/ethnicity, income, role, etc differed among these group or not. We did determine that there was a relationship between a student's job status and their satisfaction with the variety of produce offered. Respondents believe that the market provides an inclusive and welcoming space, but some were not as satisfied with accessibility. Faculty and staff were generally less satisfied in this area than students. A possible reason for this difference might be that faculty and staff are older and therefore more likely to have mobility issues that may make it difficult for them to attend the market at its current location.

For respondents who have not been to the market, most said they could not attend due to the time. Since the market is only open for a short time on Fridays, changing the time or extending hours might increase the number of attendees. Students in particular have issues with the time, but they are also less likely to need the produce anyway because a majority of them have meal plans. The second most common reason for not attending the market was that it was at bad location. While the current location is certainly convenient for the volunteers and other farm staff, it seems that it is not nearly as convenient for Saint Mary's community members. Moving the market to somewhere more centralized such as in front of the Student Center could increase the number of attendees. Another option would be to keep the current location, but set up a small table in a more central location with less variety and/or quantity of produce than the main location. This option would reduce the amount of produce that would need to be transported from the farm to the central location while still allowing community members who struggle with the current location to have an opportunity to attend. Changing the location might also make it easier for people who struggle with the time to attend because they would not have to spend as much time walking to the market.

No respondents said they were unaware of the market happening, so current advertising efforts are effective. The most common way respondents have heard of the market was through the emails, and word of mouth was the second most common. Social media was the least common way of hearing about the farm, so future marketing strategies could focus on expanding social media presence. Social media would be particularly important for increasing the number of attendees who exist outside the Saint Mary's community such as South Bend residents.

Most people do not believe that the market is an important resource for them personally, and they do not get a majority of their produce from the market. This result makes sense because a third of the respondents are students who very often have meal plans, so they do not have as much need for the produce as other community members might. Additionally, faculty and staff who make up the other two-thirds of respondents have a steady job at the college, so even though they may or may not be in one of the lower income categories, they most likely have the ability to pay for produce at the grocery store. Students are typically more comfortable taking produce without paying than faculty or staff which is probably because students are more used to getting free things on campus.

Our analysis could not provide as much insight as we would have liked. Due to the majority of the Saint Mary's population being white, it was difficult to glean any real results relating to how satisfaction differs between people of various ethnicities and cultural backgrounds. Additionally, for some of the questions, there were not enough responses in general to allow us to accurately test if income, student job status, or role affected how people responded. If the survey had been allowed to run for a longer period of time, we could have collected more responses, thus increasing the sample size. Also, distributing the survey to people who are outside the Saint Mary's community could have provided more insight into how the market could better serve visitors who are not affiliated with the college and would likely increase the diversity in cultural background and income represented in the survey. Future data collection could focus on this group of people to learn how the market could better serve them.

Overall, our analysis showed that the market is serving the Saint Mary's community well but with a few areas for improvement. The Saint Mary's community would like to see an increase in the variety and quantity of produce available, and they would like the location and time to be more convenient.

0.6 Conclusion

In general, the market is a good resource for the community, and most people view it in a positive light with high satisfaction levels. Increasing the variety and quantity of the produce offered at the market would help to increase satisfaction levels, and changing the location or time could improve accessibility. Additionally, future data collection with a focus on the St. Joe County community as a whole could provide better insight into how to better serve people of more diverse cultural backgrounds and socioeconomic statuses. While awareness of the market is high among members of the Saint Mary's Community, using social media as a marketing tool could help increase awareness outside of the college. Focusing on improving these areas will allow the farm to better meet their goals of creating an inclusive and welcoming space for a diverse set of individuals.

0.7 Technical Appendix

```r
data <- read.csv('CleanedDataRaceIncome.csv')

#lapply(data, table)

role <- data$Q1 # role-student, faculty, staff: coded
role1 <- role <- factor(role, levels = c("Faculty", "Staff", "Off-campus student (commuter)",
                                 "On-campus student (resident)"))
qualitySat <- data$Q16_1 # quality satisfaction [1,5]
amountSat <- data$Q16_2 # amount satisfaction   [1,5]
```

```r
varietySat <- data$Q16_3 # variety satisfaction [1,5]

income <- data$Q5 # income of faculty/staff: coded
mealPlan <- data$Q3 # meal plan for students: coded
studentJob <- data$StudentJobStatus # student jobs on/off campus: coded

raceEthnic <- data$Race_Ethnicity # race/ethnicity of people: coded

## Market Feeling Satisfaction Q22 for users: Rankings values [1,5]
welcome <- data$Q22_1
safety <- data$Q22_2
inclusion <- data$Q22_3
accessible <- data$Q22_4
fosterInc <- data$Q22_5
impactSMC_A <- data$Q22_6
impactMe <- data$Q22_7
resourceCom_A <- data$Q22_8
resourceMe <- data$Q22_9

## Market affect for non attendees
impactSMC_NA <- data$Q27_1 #agree <-> disagree [1,5]
impactCom_NA <- data$Q27_2 #         "

## Have you heard of market
poster <- data$Q10_1
email <- data$Q10_2
socialMedia <- data$Q10_3
wordMouth <- data$Q10_4
otherHeard <- data$Q10_5

## Portion of Produce for Household
portion <- data$Q14

## Comfort taking produce without paying
comfortLevel <- data$Q18

## Ability to Attend  ("Other" responses in TextResponses)
attendUnaware <- data$Q26_1
attendBadTime <- data$Q26_2
attendNotInterested <- data$Q26_3
attendBadPlace <- data$Q26_4

## Produce Interests
vegetables <- data$Q12_1
fruits <- data$Q12_2
flowers <- data$Q12_3
herbs <- data$Q12_4
noneInterest <- data$Q12_5

## Ranking of importance for produce
qualityRank <- data$Q15_1
amountRank <- data$Q15_2
varietyRank <- data$Q15_3
```

```r
roleDF <- data.frame(table(role))

ggplot(roleDF, aes(x = "", y = Freq ,fill = role)) +
  geom_col(color = "gray40") +
  coord_polar(theta = "y") +
  geom_text(aes(label = Freq),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Community Role Breakdown of Survey Responses", fill = "Community Role", x = "", y = "")+
  scale_fill_discrete(labels = c('Faculty (36)', 'Off-campus Student (commuter) (8)', 'On-campus Studen
  theme_void()

#table(raceEthnic)

counts = data.frame("Race" = c('Asian', 'Black or African American', 'Hispanic, Latino, or Spanish origi
                    "counts" = c(2, 1, 2, 1, 10, 3, 81))

ggplot(counts, aes(x = "", y = counts ,fill = Race)) +
  geom_col(color = "gray40") +
  coord_polar(theta = "y") +
  geom_text(aes(label = counts),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Race/Ethnicity Breakdown of Survey Responses", fill = "Race/Ethnicity", x = "", y = "")+
  scale_fill_discrete(labels = c('Asian (2)', 'Black or African American (1)', 'Hispanic, Latino, or Spa
  theme_void()
```

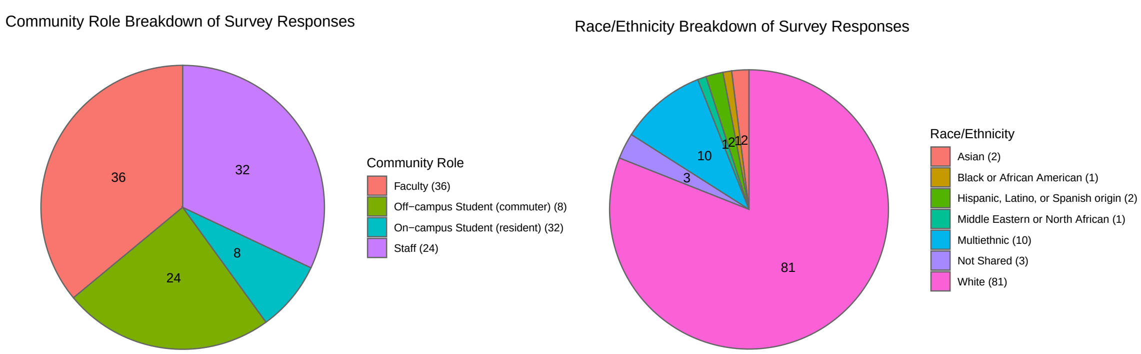

Figure 22: Community Role and Race/Ethnicity Breakdown

```r
incomeT <- table(income)
incomeCounts <- data.frame(incomeT)
incomeCounts <- incomeCounts[incomeCounts$income != "",]
incomeCounts <- incomeCounts[incomeCounts$income != 6,]


ggplot(incomeCounts, aes(x = "", y = Freq ,fill = income)) +
  geom_col(color = "gray40") +
  coord_polar(theta = "y") +
  geom_text(aes(label = Freq),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Income Breakdown of Faculty/Staff Survey Responses", fill = "Income", x = "", y = "")+
  scale_fill_discrete(labels = c('$30,001 - $60,000 (10)', '$60,001 - $90,000 (20)', '$90,0001 - $120,00
  theme_void()
```

```r
jobDF <- data.frame(table(studentJob))
jobDF <- jobDF[jobDF$studentJob != 'Not Indicated',]

ggplot(jobDF, aes(x = "", y = Freq ,fill = studentJob)) +
  geom_col(color = "gray40") +
  coord_polar(theta = "y") +
  geom_text(aes(label = Freq),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Student Job Status Breakdown of Survey Responses", fill = "Student Job Status", x = "",
  scale_fill_discrete(labels = c('Multiple Jobs (1)', 'No Job (13)', 'Off-Campus Job (5)', 'On-Campus Jo
  theme_void()

mealDF <- data.frame(table(mealPlan))

ggplot(mealDF, aes(x = "", y = Freq,fill = mealPlan)) +
  geom_col(color = "gray40") +
  coord_polar(theta = "y") +
  geom_text(aes(label = Freq),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Student Meal Plan Breakdown of Survey Responses", fill = "Do you have a meal plan?", x =
  scale_fill_discrete(labels = c('Yes (33)', 'No (6)'))+
  theme_void()
```

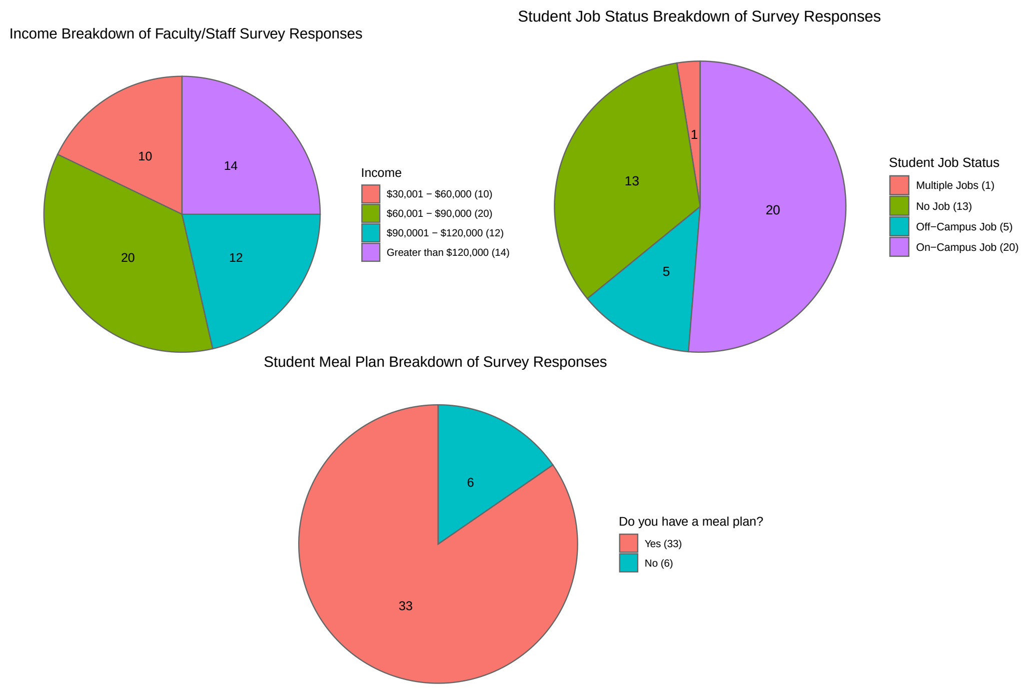

Figure 23: Socioeconomic Indicators Breakdown

```r
dfListenA <- dfListen[rowSums(dfListen) != 0,]

listen_counts <- data.frame(
  Option = colnames(dfListenA),
  Count = colSums(dfListenA)
```

```r
)

# Define the mapping of question codes to descriptive labels
listen_labels <- c(
  poster = "Poster",
  email = "Email",
  socialMedia = "Social Media",
  wordMouth = "Word of Mouth"
)

# Replace the codes with descriptive labels
listen_counts$Option <- listen_labels[listen_counts$Option]


listen_counts$Option <- factor(listen_counts$Option, levels = listen_labels)

# Create a horizontal bar plot with labels on the bars
ggplot(listen_counts, aes(x = Option, y = Count)) +
  geom_bar(stat = "identity", fill = "aquamarine4") +  # Use 'identity' for precomputed counts
  geom_text(aes(label = Count), hjust = -0.2, size = 4) +  # Add labels to the bars
  labs(
    title = "Q10 How did you hear about the Market?",
    x = NULL,  # Optional: remove the x-axis label for a cleaner look
    y = "Number of Responses",
  ) +
  scale_y_continuous(
    breaks = scales::pretty_breaks(n = 5),  # Ensure whole-number axis ticks
    expand = expansion(mult = c(0, 0.2))  # Add space for the labels
  ) +
  theme_bw() +
  theme(
    panel.grid.major.y = element_blank(),  # Remove major grid lines
    panel.grid.minor.y = element_blank(),  # Remove minor grid lines
    axis.line = element_line(color = "black"),  # Add axis lines
    axis.text.x = element_text(size = 10),  # Adjust x-axis text size
    axis.text.y = element_text(size = 10),  # Adjust y-axis text size
    plot.title = element_text(size = 12, face = "bold")  # Center and style the title
  ) +
  coord_flip()  # Flip the coordinates for horizontal bars
```

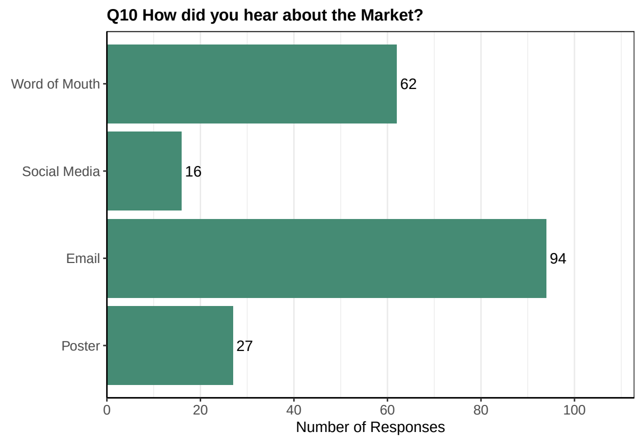

Figure 24: How People Have Heard of Market

```
# Create the stacked bar plot using ggplot2
ggplot(df1_long, aes(x = role1, y = count, fill = channel)) +
  geom_bar(stat = "identity") +  # Stacked bars
  labs(title = "How People Heard About Market By Community Role",
       x = "Role",
       y = "Count",
       fill = "Communication Method") +
  scale_fill_discrete(labels = c("Email", "Poster","Social Media", "Word of Mouth"))+
  theme(axis.text.x = element_text(angle = 15, hjust = 1))  # Rotate x-axis labels for better readabili
```

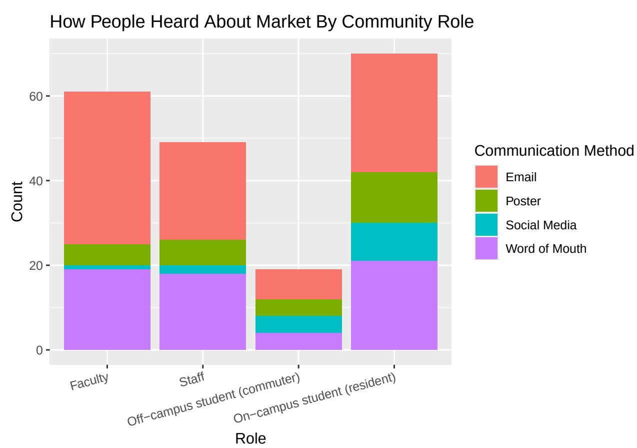

Figure 25: How People Heard About the Market Sorted by Role

```
tableCount <- table(df1_long$role1, df1_long$channel)
chiTestComMeth<- chisq.test(tableCount, simulate.p.value = TRUE, B = 10^5-1) # violates independent ass

# Additionally, explore the distribution of multiple responses:
dfListen_rm0row <- dfListen[-c(30),] # this is to remove the row of no responses at all
dfListen_rm0row$response_count <- rowSums(dfListen_rm0row)
response_distribution <- table(dfListen_rm0row$response_count)

# Create a new column for response combinations
dfListen_rm0row$response_combination <- apply(dfListen_rm0row[, 1:4], 1, function(x) {
```

```r
  paste(names(dfListen_rm0row)[1:4][x == 1], collapse = " + ")
})

# Summarize the combinations
combination_summary <- table(dfListen_rm0row$response_combination)
```

0.7.0.1 1.Table of All Combinations For Hearing about Market

Combination	Count
Email	31
Email + Social Media	1
Email + Social Media + Word of Mouth	3
Email + Word of Mouth	34
Poster	1
Poster + Email	2
Poster + Email + Social Media	2
Poster + Email + Social Media + Word of Mouth	9
Poster + Email + Word of Mouth	12
Poster + Social Media + Word of Mouth	1
Word of Mouth	3

```r
## Filtered out word of mouth because it was the main contributor to two response answers, now its most
#remove word of mouth as an option and do chisq test again
# Create the data frame
dfListen2 <- data.frame(poster, email, socialMedia)

# Replace NA values with 0
dfListen2[is.na(dfListen2)] <- 0

# Create the 'df1'data frame
dfList2 <- data.frame(role1, dfListen2)

# Reshape the data from wide to long format using base R reshape() function
dfList2_long <- reshape(dfList2,
                    varying = c("poster", "email", "socialMedia"),
                    v.names = "count",
                    timevar = "channel",
                    times = c("poster", "email", "socialMedia"),
                    direction = "long")

# Exclude zeros (filter out rows where count is 0)
dfList2_long <- dfList2_long[dfList2_long$count > 0, ]

ggplot(dfList2_long, aes(x = role1, y = count, fill = channel)) +
  geom_bar(stat = "identity") +  # Stacked bars
  labs(title = "How People Have Heard About Market By Community Role Without Word of Mouth",x = "Role",
  scale_fill_discrete(labels = c("Email", "Poster", "Word of Mouth"))+
  theme(axis.text.x = element_text(angle = 15, hjust = 1))  # Rotate x-axis labels for better readabili
```

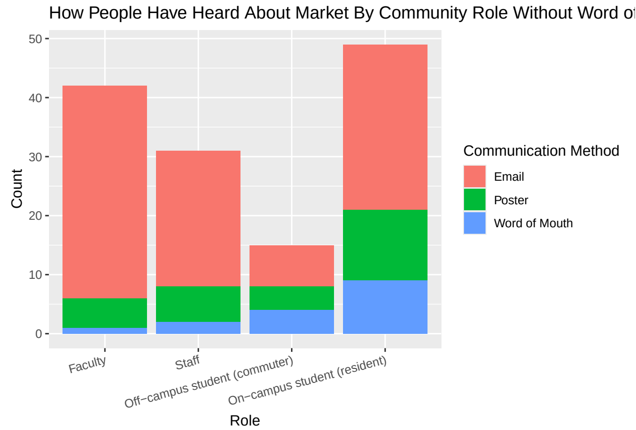

Figure 26: How People Heard About the Market Sorted by Role, Excluding by Word of Mouth

```
tableCount2 <- table(dfList2_long$role1, dfList2_long$channel)
chiTest2Com <- chisq.test(tableCount2, simulate.p.value = TRUE, B = 10^5-1) # violates independent assu
```

```
# Calculate conditional probabilities (proportions)
proportions_table2 <- proportions(tableCount2, 1)  # Conditional probabilities by rows
proportions_table2
```

```
##
##                                      email     poster socialMedia
##   Faculty                       0.85714286 0.11904762  0.02380952
##   Staff                         0.74193548 0.19354839  0.06451613
##   Off-campus student (commuter) 0.46666667 0.26666667  0.26666667
##   On-campus student (resident)  0.57142857 0.24489796  0.18367347
```

```
specialdata <- read.csv('CleanedDataRaceIncomeProp.csv')
proportion <- data.frame(specialdata$Q14)

proportion <- proportion[proportion != '',]
proportion <- data.frame(proportion)
```

```
proportion$proportion <- factor(proportion$proportion, levels = c('< 25%','25-50%', 'About 50%', '50-75
ggplot(proportion, aes(x = proportion))+
  geom_bar(fill = 'aquamarine4')+
  labs(title = str_wrap('What proportion of your household produce comes from the Market?', 60), x = 'P
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    panel.grid.major.y = element_blank(),
    panel.grid.minor.y = element_blank(),
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold")
  )
```

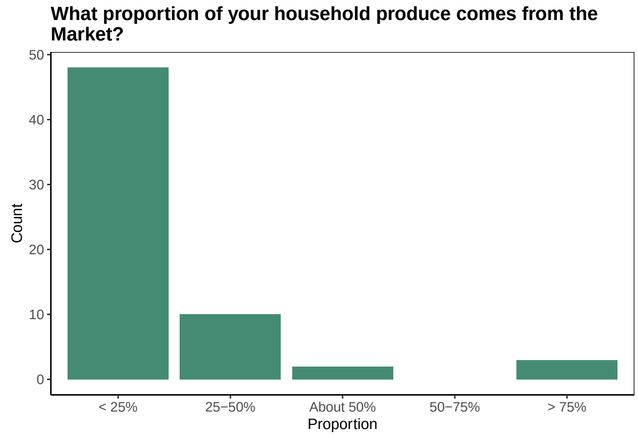

Figure 27: Proportion of Household Produce Coming From the Market for General Public

```r
# Combine role and portion into a data frame and remove rows with NA values
df2 <- na.omit(data.frame(role1, portion))

# Create a contingency table of counts
table <- table(df2$role1, df2$portion)

# Convert the contingency table into a data frame for ggplot
tableDF <- as.data.frame(table)
colnames(tableDF) <- c("role", "portion_of_Household_Produce", "count")

# Create bar plot for counts of each combination of role and portion
ggplot(tableDF, aes(x = role, y = count, fill = portion_of_Household_Produce)) +
  geom_bar(stat = "identity") +
  scale_fill_discrete(name = "Proportion of Household Produce from Market",
                      labels = c("Less than 25%", "25%-50%", "About 50%", "More than 75%"), drop = FALSE
  scale_x_discrete(guide = guide_axis(angle = 15), drop = FALSE) +
  ggtitle("Distribution of Proportion of Household Produce from Market by Role") +
  xlab("Role") +
  ylab("Counts")
```

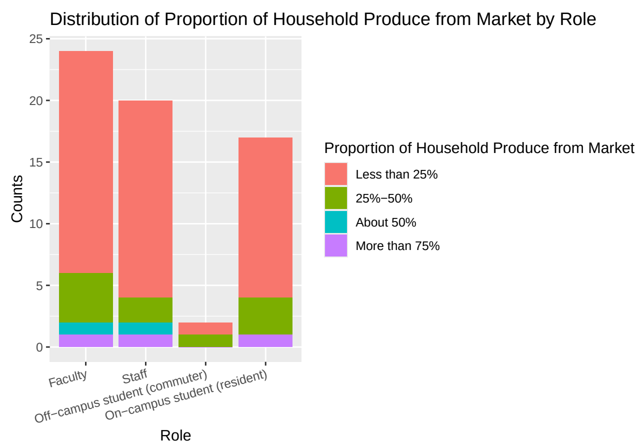

Figure 28: Proportion of Household Produce Coming From the Market Sorted by Role

```
# Calculate conditional probabilities (proportions)
proportions_table <- proportions(table, 1)  # Conditional probabilities by rows

# Convert proportions table to a data frame for ggplot
proportionsDF <- as.data.frame(as.table(proportions_table))
colnames(proportionsDF) <- c("Role", "Proportion_of_goods", "Proportion")

# Create bar plot for conditional proportions
ggplot(proportionsDF, aes(x = Role, y = Proportion, fill = Proportion_of_goods)) +
  geom_bar(stat = "identity") +
  scale_fill_discrete(name = "Portion of Household Produce from Market",
                      labels = c("Less than 25%", "25%-50%", "About 50%", "More than 75%")) +
  scale_x_discrete(guide = guide_axis(angle = 15)) +
  ggtitle("Conditional Distribution of Household Produce Portion from Market by Role") +
  xlab("Role") +
  ylab("Proportions")
```

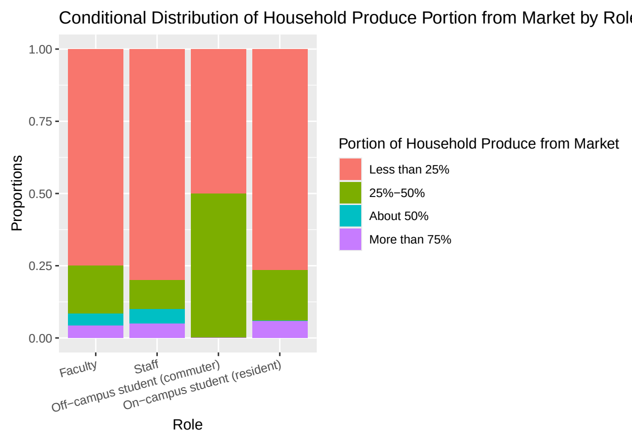

Figure 29: Proportion Breakdown of Proportion of Household Produce Coming From Market by Role

```
tablePortion <- table(df2$role, df2$portion)
chiTestPortion <- chisq.test(tablePortion, simulate.p.value = TRUE, B = 10^5-1)
```

```
incomeProp <- data.frame(income, portion)
incomeProp <- incomeProp[!(is.na(incomeProp$portion)),]
incomeProp <- incomeProp[incomeProp$income != '',]
incomeProp <- incomeProp[incomeProp$income != '6',]

ggplot(incomeProp, aes(x = portion, fill = income))+
  geom_bar()+
  scale_fill_discrete(drop = FALSE)+
  scale_x_discrete(drop = FALSE)+
  labs(title = "Proportion of Household Produce recieved from Market by Income", x = "Proportion of Proc
  theme_bw()
```

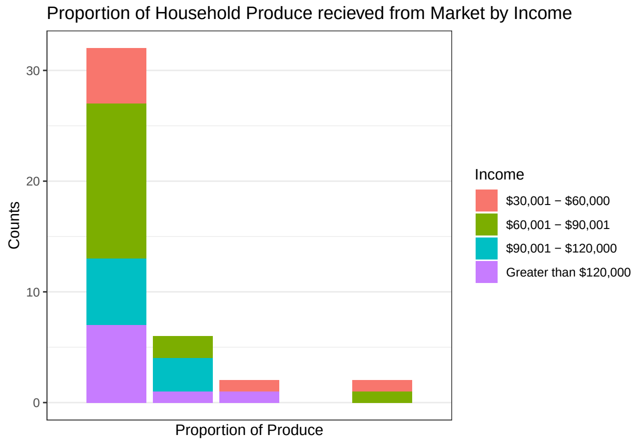

Figure 30: Proportion of Household Produce recieved from the Market by Income

```
incomePropTab <- table(income, portion)
chisq.test(incomePropTab, simulate.p.value = TRUE, B = 10^5)
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  incomePropTab
## X-squared = 12.101, df = NA, p-value = 0.5892
```

```
satisfaction <- data.frame(qualitySat, varietySat, amountSat)
satis_responses <- satisfaction[rowSums(is.na(satisfaction)) < ncol(satisfaction),]

ggplot(satis_responses, aes(x = qualitySat)) +
  geom_bar(fill = "aquamarine4") +
  labs(title = "Satisfaction with Quality of Produce", x = "Ranking", y = "Count") +
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(satis_responses, aes(x = varietySat)) +
  geom_bar(fill = "aquamarine4") +
  labs(title = "Satisfaction with Variety of Produce", x = "Ranking", y = "Count") +
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
```

```
  )

ggplot(satis_responses, aes(x = amountSat)) +
  geom_bar(fill = "aquamarine4") +
  labs(title = "Satisfaction with Amount of Produce", x = "Ranking", y = "Count") +
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

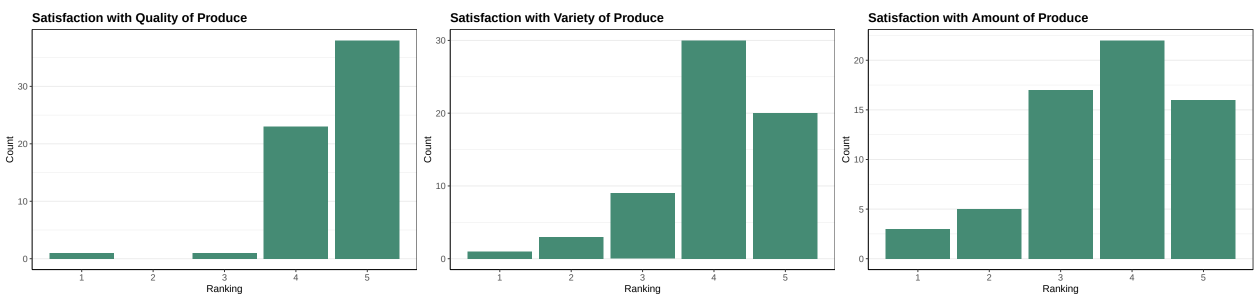

Figure 31: Satisfaction Level [1-5] For Categories of Product Attributes by General Public

```
role <- data[rownames(satis_responses), "Q1"]
satis_role <- data.frame(role, satis_responses)

ggplot(satis_role, aes(x = qualitySat, fill = role)) +
  geom_bar() +
  labs(title = "Satisfaction with Quality of Produce by Community Role", x = "Ranking", y = "Count", fil
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(satis_role, aes(x = varietySat, fill = role)) +
  geom_bar() +
  labs(title = "Satisfaction with Variety of Produce by Community Role", x = "Ranking", y = "Count", fil
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
```

```r
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(satis_role, aes(x = amountSat, fill = role)) +
  geom_bar() +
  labs(title = "Satisfaction with Amount of Produce by Community Role", x = "Ranking", y = "Count", fil
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

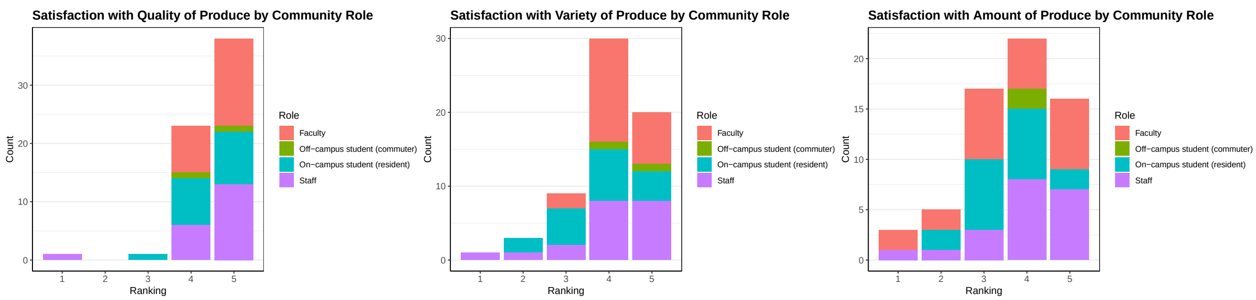

Figure 32: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Role

```r
qualityXRoleTab <- table(role1, qualitySat); addmargins(qualityXRoleTab)
```

```
##                                qualitySat
## role1                            1  3  4  5 Sum
##   Faculty                        0  0  8 15  23
##   Staff                          1  0  6 13  20
##   Off-campus student (commuter)  0  0  1  1   2
##   On-campus student (resident)   0  1  8  9  18
##   Sum                            1  1 23 38  63
```

```r
varietyXRoleTab <- table(role1, varietySat); addmargins(varietyXRoleTab)
```

```
##                                varietySat
## role1                            1  2  3  4  5 Sum
##   Faculty                        0  0  2 14  7  23
##   Staff                          1  1  2  8  8  20
##   Off-campus student (commuter)  0  0  0  1  1   2
##   On-campus student (resident)   0  2  5  7  4  18
##   Sum                            1  3  9 30 20  63
```

```r
amountXRoleTab <- table(role1, amountSat); addmargins(amountXRoleTab)
```

```
##                                amountSat
## role1                            1  2  3  4  5 Sum
##   Faculty                        2  2  7  5  7  23
##   Staff                          1  1  3  8  7  20
##   Off-campus student (commuter)  0  0  0  2  0   2
```

```
##   On-campus student (resident)   0  2  7  7  2  18
##   Sum                            3  5 17 22 16  63
```

```r
c2Quality <- chisq.test(qualityXRoleTab, simulate.p.value = TRUE, B = 10^5); c2Quality
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  qualityXRoleTab
## X-squared = 5.8283, df = NA, p-value = 0.7252
```

```r
c2Variety <- chisq.test(varietyXRoleTab, simulate.p.value = TRUE, B = 10^5); c2Variety
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  varietyXRoleTab
## X-squared = 10.754, df = NA, p-value = 0.4862
```

```r
c2Amount <- chisq.test(amountXRoleTab, simulate.p.value = TRUE, B = 10^5); c2Amount
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  amountXRoleTab
## X-squared = 11.684, df = NA, p-value = 0.4875
```

```r
race <- data[rownames(satis_responses), "Race_Ethnicity"]
satis_race <- data.frame(race, satis_responses)

ggplot(satis_race, aes(x = qualitySat, fill = race)) +
  geom_bar() +
  labs(title = "Satisfaction with Quality of Produce by Ethnicity", x = "Ranking", y = "Count", fill =
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
ggplot(satis_race, aes(x = varietySat, fill = race)) +
  geom_bar() +
  labs(title = "Satisfaction with Variety of Produce by Ethnicity", x = "Ranking", y = "Count", fill =
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
```

```r
)

ggplot(satis_race, aes(x = amountSat, fill = race)) +
  geom_bar() +
  labs(title = "Satisfaction with Amount of Produce by Ethnicity", x = "Ranking", y = "Count", fill = "
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

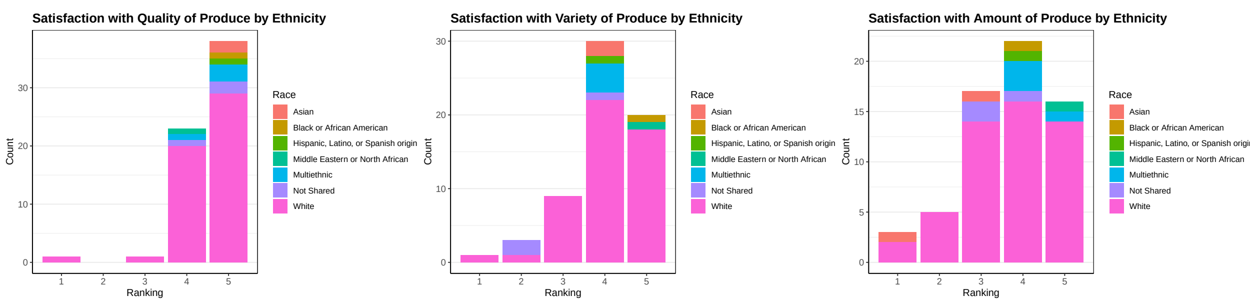

Figure 33: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Race and Ethnicity

```r
qualityXRaceTab <- table(raceEthnic, qualitySat); addmargins(qualityXRaceTab)
```

```
##                                      qualitySat
## raceEthnic                             1  3  4  5 Sum
##   Asian                                0  0  0  2   2
##   Black or African American            0  0  0  1   1
##   Hispanic, Latino, or Spanish origin  0  0  0  1   1
##   Middle Eastern or North African      0  0  1  0   1
##   Multiethnic                          0  0  1  3   4
##   Not Shared                           0  0  1  2   3
##   White                                1  1 20 29  51
##   Sum                                  1  1 23 38  63
```

```r
varietyXRaceTab <- table(raceEthnic, varietySat); addmargins(varietyXRaceTab)
```

```
##                                      varietySat
## raceEthnic                             1  2  3  4  5 Sum
##   Asian                                0  0  0  2  0   2
##   Black or African American            0  0  0  0  1   1
##   Hispanic, Latino, or Spanish origin  0  0  0  1  0   1
##   Middle Eastern or North African      0  0  0  0  1   1
##   Multiethnic                          0  0  0  4  0   4
##   Not Shared                           0  2  0  1  0   3
##   White                                1  1  9 22 18  51
##   Sum                                  1  3  9 30 20  63
```

```r
amountXRaceTab <- table(raceEthnic, amountSat); addmargins(amountXRaceTab)
```

```
##                                       amountSat
## raceEthnic                              1  2  3  4  5 Sum
##   Asian                                 1  0  1  0  0   2
##   Black or African American             0  0  0  1  0   1
##   Hispanic, Latino, or Spanish origin   0  0  0  1  0   1
##   Middle Eastern or North African       0  0  0  0  1   1
##   Multiethnic                           0  0  0  3  1   4
##   Not Shared                            0  0  2  1  0   3
##   White                                 2  5 14 16 14  51
##   Sum                                   3  5 17 22 16  63
```

```r
c2QualityR <- chisq.test(qualityXRaceTab, simulate.p.value = TRUE, B = 10^5); c2QualityR
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  qualityXRaceTab
## X-squared = 5.2023, df = NA, p-value = 0.8101
```

```r
c2VarietyR <- chisq.test(varietyXRaceTab, simulate.p.value = TRUE, B = 10^5); c2VarietyR
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  varietyXRaceTab
## X-squared = 39.406, df = NA, p-value = 0.1241
```

```r
c2AmountR <- chisq.test(amountXRaceTab, simulate.p.value = TRUE, B = 10^5); c2AmountR
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  amountXRaceTab
## X-squared = 23.913, df = NA, p-value = 0.4279
```

```r
facStaffSat <- satis_role[satis_role$role == 'Faculty'|satis_role$role == 'Staff',]

income <- data[rownames(facStaffSat), "Q5"]
facStaffIncomeSat <- data.frame(income, facStaffSat)

facStaffIncomeSat <- facStaffIncomeSat[facStaffIncomeSat$income != '6.0', ]
facStaffIncomeSat <- facStaffIncomeSat[facStaffIncomeSat$income != '', ]

facStaffIncomeSat$qualitySat <- factor(facStaffIncomeSat$qualitySat, levels = c(1,2,3,4,5))
ggplot(facStaffIncomeSat, aes(x = qualitySat, fill = income)) +
  geom_bar() +
  labs(title = "Satisfaction with Quality of Produce by Income", x = "Ranking", y = "Count", fill = "In
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  theme_bw()+
  theme(
```

```
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

facStaffIncomeSat$varietySat <- factor(facStaffIncomeSat$varietySat, levels = c(1,2,3,4,5))
ggplot(facStaffIncomeSat, aes(x = varietySat, fill = income)) +
  geom_bar() +
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  labs(title = "Satisfaction with Variety of Produce by Income", x = "Ranking", y = "Count", fill = "Inc
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(facStaffIncomeSat, aes(x = amountSat, fill = income)) +
  geom_bar() +
  labs(title = "Satisfaction with Amount of Produce by Income", x = "Ranking", y = "Count", fill = "Inco
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

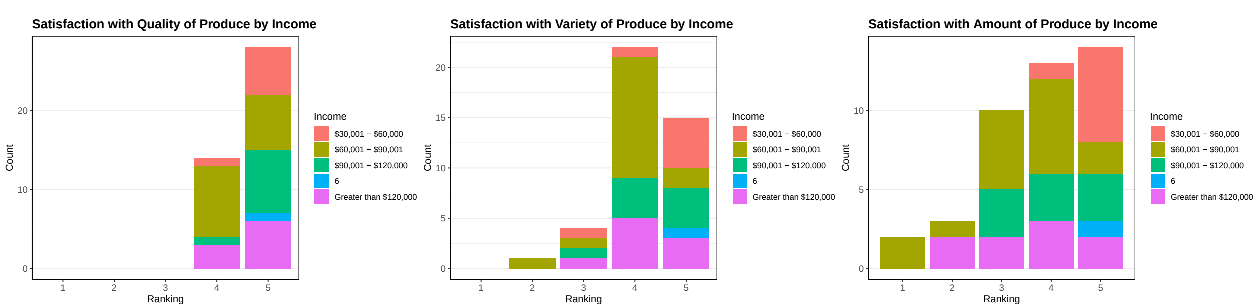

Figure 34: Satisfaction Level [1-5] For Categories of Product Attributes Sorted by Income

```
qualityXIncomeTab <- table(facStaffIncomeSat$income, facStaffIncomeSat$qualitySat); addmargins(qualityX
```

```
##
##                            1   2   3   4   5 Sum
```

```
##   $30,001 - $60,000      0  0  0  1  6   7
##   $60,001 - $90,001      0  0  0  9  7  16
##   $90,001 - $120,000     0  0  0  1  8   9
##   6                      0  0  0  0  1   1
##   Greater than $120,000  0  0  0  3  6   9
##   Sum                    0  0  0 14 28  42
```

```
varietyXIncomeTab <- table(facStaffIncomeSat$income, facStaffIncomeSat$varietySat); addmargins(varietyXI
```

```
##
##                          1  2  3  4  5 Sum
##   $30,001 - $60,000      0  0  1  1  5   7
##   $60,001 - $90,001      0  1  1 12  2  16
##   $90,001 - $120,000     0  0  1  4  4   9
##   6                      0  0  0  0  1   1
##   Greater than $120,000  0  0  1  5  3   9
##   Sum                    0  1  4 22 15  42
```

```
amountXIncomeTab <- table(facStaffIncomeSat$income, facStaffIncomeSat$amountSat); addmargins(amountXInc
```

```
##
##                          1  2  3  4  5 Sum
##   $30,001 - $60,000      0  0  0  1  6   7
##   $60,001 - $90,001      2  1  5  6  2  16
##   $90,001 - $120,000     0  0  3  3  3   9
##   6                      0  0  0  0  1   1
##   Greater than $120,000  0  2  2  3  2   9
##   Sum                    2  3 10 13 14  42
```

```
c2QualityI <- chisq.test(qualityXIncomeTab, simulate.p.value = TRUE, B = 10^5); c2QualityI
```

```
## Warning in chisq.test(qualityXIncomeTab, simulate.p.value = TRUE, B = 10^5):
## cannot compute simulated p-value with zero marginals

## Warning in chisq.test(qualityXIncomeTab, simulate.p.value = TRUE, B = 10^5):
## Chi-squared approximation may be incorrect

##
##  Pearson's Chi-squared test
##
## data:  qualityXIncomeTab
## X-squared = NaN, df = 16, p-value = NA
```

```
c2VarietyI <- chisq.test(varietyXIncomeTab, simulate.p.value = TRUE, B = 10^5); c2VarietyI
```

```
## Warning in chisq.test(varietyXIncomeTab, simulate.p.value = TRUE, B = 10^5):
## cannot compute simulated p-value with zero marginals

## Warning in chisq.test(varietyXIncomeTab, simulate.p.value = TRUE, B = 10^5):
## Chi-squared approximation may be incorrect

##
##  Pearson's Chi-squared test
##
## data:  varietyXIncomeTab
## X-squared = NaN, df = 16, p-value = NA
```

```
c2AmountI <- chisq.test(amountXIncomeTab, simulate.p.value = TRUE, B = 10^5); c2AmountI
```

```
##
```

```
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  amountXIncomeTab
## X-squared = 20.681, df = NA, p-value = 0.2013
```

```
studentSat <- satis_role[satis_role$role == "On-campus student (resident)"|
                           satis_role$role == "Off-campus student (commuter)",]
jobStat <- data[rownames(studentSat), "StudentJobStatus"]
studentJobSat <- data.frame(jobStat, studentSat)

studentJobSat$qualitySat <- factor(studentJobSat$qualitySat, levels = c(2,3,4,5))
ggplot(studentJobSat, aes(x = qualitySat, fill = jobStat)) +
  geom_bar() +
  labs(title = "Satisfaction with Quality of Produce by Student Job Status", x = "Ranking", y = "Count"
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

studentJobSat$varietySat <- factor(studentJobSat$varietySat, levels = c(2,3,4,5))
ggplot(studentJobSat, aes(x = varietySat, fill = jobStat)) +
  geom_bar() +
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  labs(title = "Satisfaction with Variety of Produce by Student Job Status", x = "Ranking", y = "Count"
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

studentJobSat$amountSat <- factor(studentJobSat$amountSat, levels = c(2,3,4,5))
ggplot(studentJobSat, aes(x = amountSat, fill = jobStat)) +
  geom_bar() +
  scale_x_discrete(drop = FALSE) +
  scale_fill_discrete(drop = FALSE) +
  labs(title = "Satisfaction with Amount of Produce by Student Job Status", x = "Ranking", y = "Count",
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
```

```
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

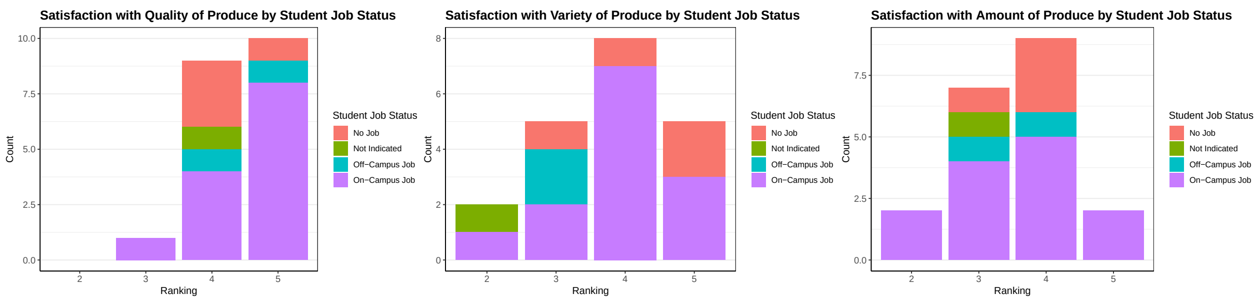

Figure 35: Satisfaction Level [1-5] For Categories of Product Attributes Sorted Job Status

```
qualityXJobTab <- table(studentJobSat$jobStat, studentJobSat$qualitySat); addmargins(qualityXJobTab)
```

```
##
##                   2  3  4  5 Sum
##   No Job          0  0  3  1   4
##   Not Indicated   0  0  1  0   1
##   Off-Campus Job  0  0  1  1   2
##   On-Campus Job   0  1  4  8  13
##   Sum             0  1  9 10  20
```

```
varietyXJobTab <- table(studentJobSat$jobStat, studentJobSat$varietySat); addmargins(varietyXJobTab)
```

```
##
##                   2  3  4  5 Sum
##   No Job          0  1  1  2   4
##   Not Indicated   1  0  0  0   1
##   Off-Campus Job  0  2  0  0   2
##   On-Campus Job   1  2  7  3  13
##   Sum             2  5  8  5  20
```

```
amountXJobTab <- table(studentJobSat$jobStat, studentJobSat$amountSat); addmargins(amountXJobTab)
```

```
##
##                   2  3  4  5 Sum
##   No Job          0  1  3  0   4
##   Not Indicated   0  1  0  0   1
##   Off-Campus Job  0  1  1  0   2
##   On-Campus Job   2  4  5  2  13
##   Sum             2  7  9  2  20
```

```
c2QualityJ <- chisq.test(qualityXJobTab, simulate.p.value = TRUE, B = 10^5); c2QualityJ
```

```
## Warning in chisq.test(qualityXJobTab, simulate.p.value = TRUE, B = 10^5):
## cannot compute simulated p-value with zero marginals

## Warning in chisq.test(qualityXJobTab, simulate.p.value = TRUE, B = 10^5):
## Chi-squared approximation may be incorrect

##
```

```
##  Pearson's Chi-squared test
##
## data:  qualityXJobTab
## X-squared = NaN, df = 9, p-value = NA
```

```r
c2VarietyJ <- chisq.test(varietyXJobTab, simulate.p.value = TRUE, B = 10^5); c2VarietyJ
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  varietyXJobTab
## X-squared = 17.817, df = NA, p-value = 0.03064
```

```r
c2AmountJ <- chisq.test(amountXJobTab, simulate.p.value = TRUE, B = 10^5); c2AmountJ
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  amountXJobTab
## X-squared = 5.0549, df = NA, p-value = 0.8622
```

```r
# Combine role and comfortLevel into a data frame and remove rows with NA values
dfComfort <- na.omit(data.frame(role1, comfortLevel))

# Create a contingency table of counts
tableComfort <- table(dfComfort$role, dfComfort$comfortLevel)

# Convert the contingency table into a data frame for ggplot
ComfortDf <- as.data.frame(tableComfort)
colnames(ComfortDf) <- c("role", "Comfortability with Not Paying", "count")

# Create a bar plot for counts of each combination of role and comfortability
ggplot(ComfortDf, aes(x = role, y = count, fill = `Comfortability with Not Paying`)) +
  geom_bar(stat = "identity") +
  scale_fill_discrete(name = "Comfortability Not Paying",
                      labels = c("Very Uncomfortable", "Uncomfortable",
                                 "Neither comfortable nor uncomfortable",
                                 "Comfortable", "Very Comfortable")) +
  scale_x_discrete(guide = guide_axis(angle = 15)) +
  ggtitle("Distribution of Comfortability Not Paying by Role") +
  xlab("Role") +
  ylab("Counts") +
  theme_bw()
```

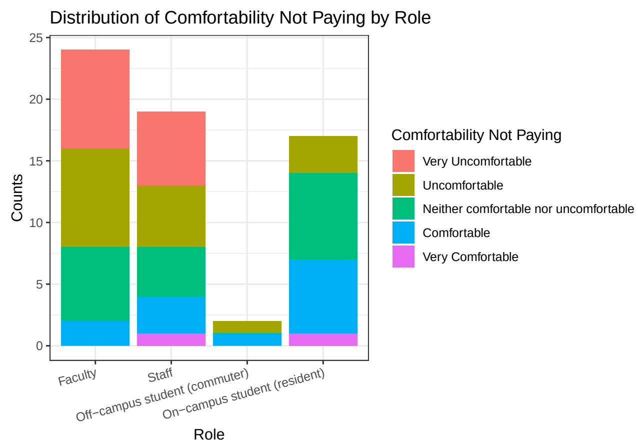

Figure 36: People's Comfort Levels Not Paying for Produce

```
proportions(tableComfort, 1)
```

```
##
##                                        1          2          3          4
##   Faculty                     0.33333333 0.33333333 0.25000000 0.08333333
##   Staff                       0.31578947 0.26315789 0.21052632 0.15789474
##   Off-campus student (commuter) 0.00000000 0.50000000 0.00000000 0.50000000
##   On-campus student (resident)  0.00000000 0.17647059 0.41176471 0.35294118
##
##                                        5
##   Faculty                     0.00000000
##   Staff                       0.05263158
##   Off-campus student (commuter) 0.00000000
##   On-campus student (resident)  0.05882353
```

```
socialJ <- data.frame(welcome, safety, inclusion, accessible, fosterInc, impactSMC_A, impactMe, resourc

socialJ_responses <- socialJ[rowSums(is.na(socialJ)) < ncol(socialJ),]

ggplot(socialJ_responses, aes(x=welcome)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "I felt welcome at the Market.", x = "Strongly Disagree <--> Strongly Agree", y = "Count
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x=safety)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "I felt safe at the Market.", x = "Strongly Disagree <--> Strongly Agree", y = "Count")
  theme_bw()+
  theme(
```

```
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x=inclusion)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "I felt included at the Market.", x = "Strongly Disagree <--> Strongly Agree", y = "Coun
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x = accessible))+
  geom_bar(fill = "aquamarine4") +
  labs(title = "The Market is accessible to me.", x = "Strongly Disagree <--> Strongly Agree", y = "Coun
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold")
  )

ggplot(socialJ_responses, aes(x=fosterInc)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "The Market fosters an inclusive environment.", x = "Strongly Disagree <--> Strongly Agre
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x = impactSMC_A))+
  geom_bar(fill = "aquamarine4")+
  labs(title = "The Market has a positive impact on the SMC community.", x = "Strongly Disagree <--> St
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
```

```
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x=impactMe)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "The Market has a postive impact on me.", x = "Strongly Disagree <--> Strongly Agree", y
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x=resourceCom_A)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "The Market is an important resource for the community.", x = "Strongly Disagree <--> St
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )

ggplot(socialJ_responses, aes(x=resourceMe)) +
  geom_bar(fill = "aquamarine4")+
  labs(title = "The Market is an important resource for me.", x = "Strongly Disagree <--> Strongly Agre
  theme_bw()+
  theme(
    panel.grid.major.x = element_blank(), # Remove major grid lines
    panel.grid.minor.x = element_blank(), # Remove minor grid lines
    axis.line = element_line(color = "black"), # Add axis lines
    axis.text.x = element_text(size = 10), # Adjust x-axis text size
    axis.text.y = element_text(size = 10), # Adjust y-axis text size
    plot.title = element_text(size = 14, face = "bold") # Center and style the title
  )
```

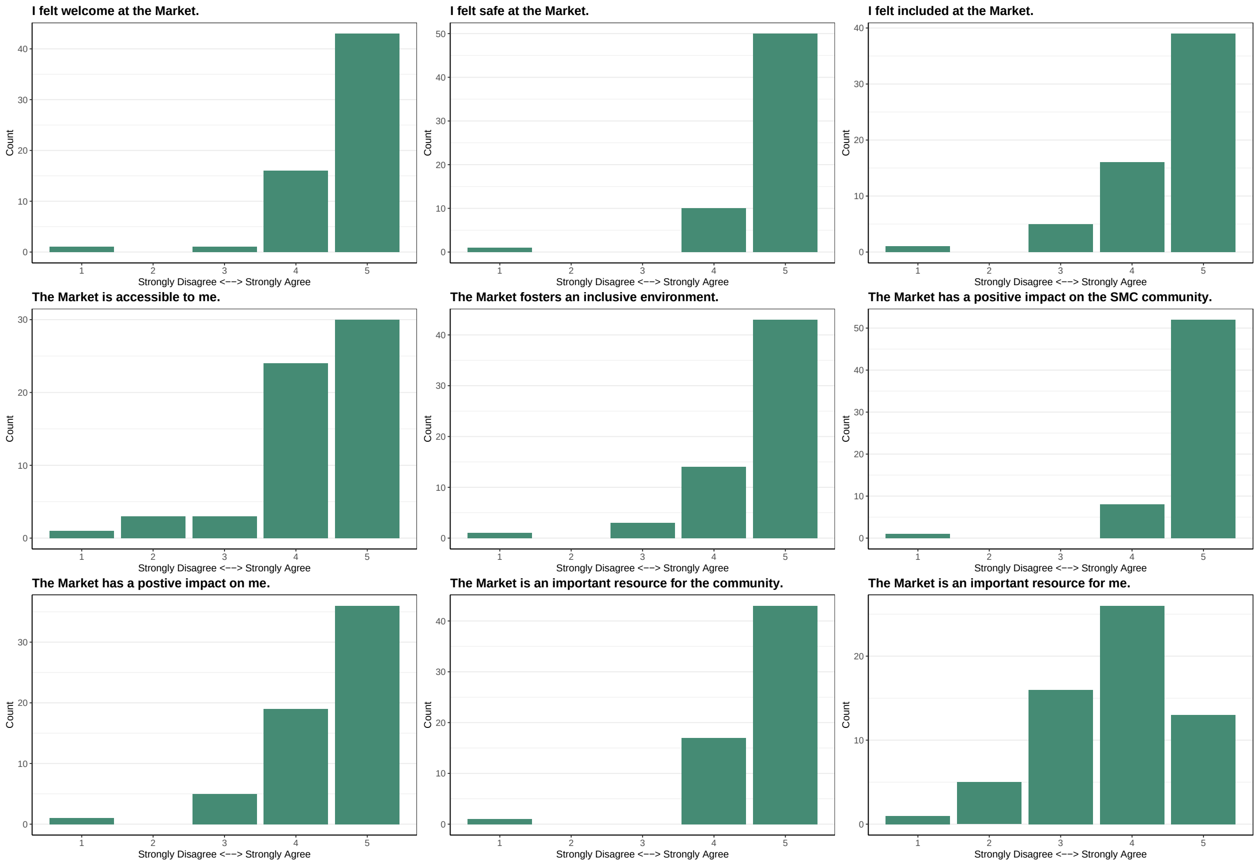

Figure 37: Satisfaction Level [1-5] on Opinion Statements of the Market's Goals by General Public

```r
## Market Feeling Satisfaction Q22 for users: Rankings values [1,5]
accessible <- data$Q22_4
impactMe <- data$Q22_7
resourceMe <- data$Q22_9

df3 <- na.omit(data.frame(role1, welcome, safety, inclusion, accessible, fosterInc, impactMe, resourceM

## median for each question by role
AccMedian <-tapply(df3$accessible, df3$role1, median) ## Accessible

# For accessibility satisfaction
df3$accessible_factor <- factor(df3$accessible, levels = c(1, 2, 3, 4, 5))
accessible_summary <- as.data.frame(table(df3$role1, df3$accessible_factor))
colnames(accessible_summary) <- c("Role", "Satisfaction_Level", "Count")

# Plot
ggplot(accessible_summary, aes(x = Satisfaction_Level, y = Count, fill = Role)) +
  geom_bar(stat = "identity") +
  labs(
    title = "'The Market is Accessible to Me' Satisfaction by Role",
    x = "Satisfaction Level",
    y = "Count",
    fill = "Role"
```

```r
  ) +
  theme(axis.text.x = element_text(angle = 15, vjust = 1, hjust=1))
```

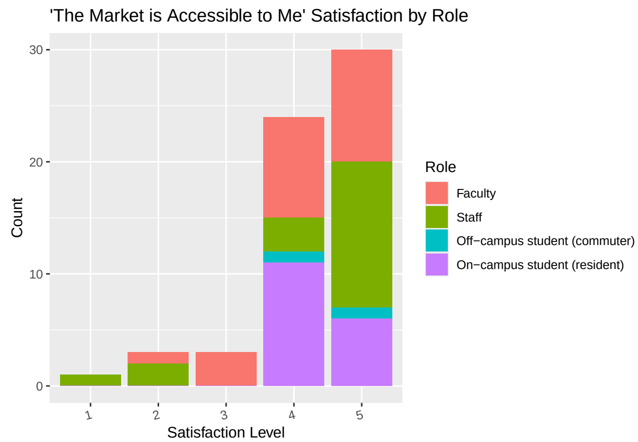

Figure 38: Satisfaction Level [1-5] For 'The Market is Accessible to Me' Statement Sorted by Roles

```r
proportions(table(df3$role1, df3$accessible), 1)
```

```
##
##                                        1          2          3          4
##   Faculty                       0.00000000 0.04347826 0.13043478 0.39130435
##   Staff                         0.05263158 0.10526316 0.00000000 0.15789474
##   Off-campus student (commuter) 0.00000000 0.00000000 0.00000000 0.50000000
##   On-campus student (resident)  0.00000000 0.00000000 0.00000000 0.64705882
##
##                                        5
##   Faculty                       0.43478261
##   Staff                         0.68421053
##   Off-campus student (commuter) 0.50000000
##   On-campus student (resident)  0.35294118
```

```r
roleAccess <- data.frame(raceEthnic, accessible)
roleAccess <- roleAccess[!(is.na(roleAccess$accessible)),]

ggplot(roleAccess, aes(x = accessible, fill = raceEthnic))+
  geom_bar()+
  labs(title = "The Market is Accessible to me by Race/Ethnicity", x = "Strongly Disagree <--> Strongly
  theme_bw()
```

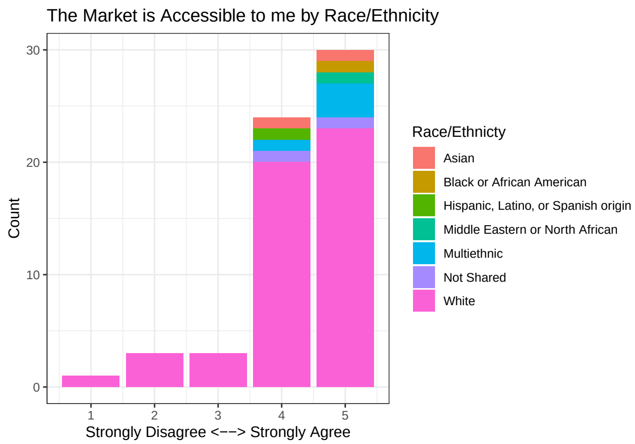

Figure 39: Agreement with the statement "The Market is Accessible to me" by Race/Ethnicity

```
raceAccessibility <- table(raceEthnic, accessible); raceAccessibility
```

```
##                                      accessible
## raceEthnic                             1  2  3  4  5
##   Asian                                0  0  0  1  1
##   Black or African American            0  0  0  0  1
##   Hispanic, Latino, or Spanish origin  0  0  0  1  0
##   Middle Eastern or North African      0  0  0  0  1
##   Multiethnic                          0  0  0  1  3
##   Not Shared                           0  0  0  1  1
##   White                                1  3  3 20 23
```

```
chisq.test(raceAccessibility, simulate.p.value = TRUE, B = 10^5)
```

```
##
##  Pearson's Chi-squared test with simulated p-value (based on 1e+05
##  replicates)
##
## data:  raceAccessibility
## X-squared = 5.7797, df = NA, p-value = 0.977
```

```
# Create the data frame
dfAttend <- data.frame(attendUnaware, attendBadTime, attendNotInterested, attendBadPlace)

# Replace NA values with 0
dfAttend[is.na(dfAttend)] <- 0

df4 <- data.frame(role1, dfAttend)

# Reshape the data from wide to long format using base R reshape() function
df4_long <- reshape(df4,
                    varying = c("attendUnaware", "attendBadTime", "attendNotInterested", "attendBadPlac
                    v.names = "count",
                    timevar = "Attendance",
                    times = c("attendUnaware", "attendBadTime", "attendNotInterested", "attendBadPlace"
                    direction = "long")

# Exclude zeros (filter out rows where count is 0)
```

```r
df4_long <- df4_long[df4_long$count > 0, ]

ggplot(df4_long, aes(x = role1, y = count, fill = Attendance)) +
  geom_bar(stat = "identity") +  # Stacked bars
  labs(title = "Why People Do not Attend Market Sorted by Community Role", x = "Role", y = "Count", fil
  scale_fill_discrete(labels = c("Bad Location", "Bad Time", "Not Interested"))+
  theme_bw()+
  theme(axis.text.x = element_text(angle = 15, hjust = 1))  # Rotate x-axis labels for better readabili
```

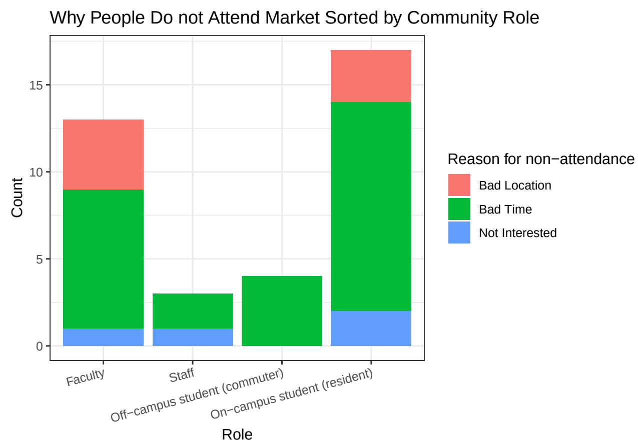

Figure 40: Why People do not Attend the Market

```r
proportions(table(df4_long$role, df4_long$Attendance), 1)
```

```
##
##                                 attendBadPlace attendBadTime
##   Faculty                           0.30769231    0.61538462
##   Staff                             0.00000000    0.66666667
##   Off-campus student (commuter)     0.00000000    1.00000000
##   On-campus student (resident)      0.17647059    0.70588235
##
##                                 attendNotInterested
##   Faculty                                0.07692308
##   Staff                                  0.33333333
##   Off-campus student (commuter)          0.00000000
##   On-campus student (resident)           0.11764706
```

```r
wants <- data.frame(vegetables, fruits, flowers, herbs, noneInterest)
wants_responses <- wants[rowSums(is.na(wants)) < ncol(wants), ]
wants_responses[is.na(wants_responses)] <- 0

option_counts <- data.frame(
  Option = colnames(wants_responses),
  Count = colSums(wants_responses)
)

# Define the mapping of question codes to descriptive labels
option_labels <- c(
  vegetables = "Vegetables",
  fruits = "Fruits",
```

```r
  flowers = "Flowers",
  herbs = "Herbs",
  noneInterest = "No Interest"
)

# Replace the codes with descriptive labels
option_counts$Option <- option_labels[option_counts$Option]

new_order <- c("No Interest",
               "Herbs",
               "Flowers",
               "Fruits",
               "Vegetables"
)

option_counts$Option <- factor(option_counts$Option, levels = new_order)

# Create a horizontal bar plot with labels on the bars
ggplot(option_counts, aes(x = Option, y = Count)) +
  geom_bar(stat = "identity", fill = "aquamarine4") +  # Use 'identity' for precomputed counts
  geom_text(aes(label = Count), hjust = -0.2, size = 4) +  # Add labels to the bars
  labs(
    title = "Q12 Which types of of produce are you interested in?",
    x = NULL,  # Optional: remove the x-axis label for a cleaner look
    y = "Number of Responses",
  ) +
  scale_y_continuous(
    breaks = scales::pretty_breaks(n = 5),  # Ensure whole-number axis ticks
    expand = expansion(mult = c(0, 0.2))  # Add space for the labels
  ) +
  theme_minimal() +
  theme(
    panel.grid.major.y = element_blank(),  # Remove major grid lines
    panel.grid.minor.y = element_blank(),  # Remove minor grid lines
    axis.line = element_line(color = "black"),  # Add axis lines
    axis.text.x = element_text(size = 10),  # Adjust x-axis text size
    axis.text.y = element_text(size = 10),  # Adjust y-axis text size
    plot.title = element_text(size = 12, face = "bold")  # Center and style the title
  ) +
  coord_flip()  # Flip the coordinates for horizontal bars
```

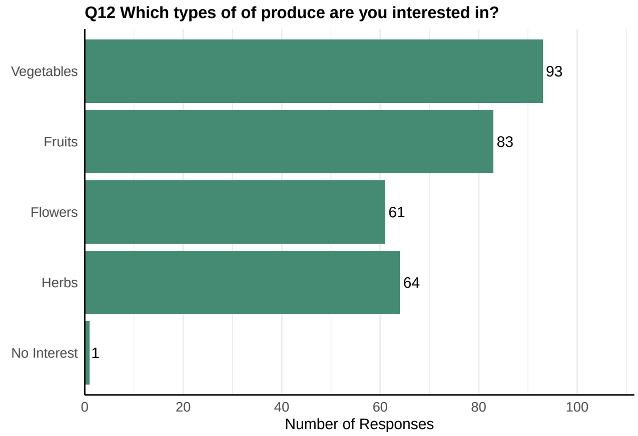

Figure 41: People's Interests in Different Type of Produce

0.7.1 Text Responses

Is there any specific produce you would like to see at the Market?

[1] "More herbs if at all possible?"
[2] "Blueberries"
[3] "Some fruit maybe"
[4] "Hot Hungarian peppers"
[5] "I would love to see more fruit!"
[6] "Peaches"
[7] "Fruit!"
[8] "Berries (if it is possible to grow them at the farm)"
[9] "I would love to see more fruits!"
[10] "no"
[11] "strawberries or raspberries!"
[12] "I'd like there to be some kind of pre-pay and pickup system rather than a free-for-all system"
[13] "Berries, beets"
[14] "African yam, sweet baby bell peppers, spinach"
[15] "more fruits such as berries and potatoes"
[16] "Green beans, cucumbers, and more squash. Would love to see some more fruit!"
[17] "fruit"
[18] "More eggs"
[19] "More Basil, Tomatoes, Eggs would be great too"
[20] "peas"
[21] "beans"
[22] "No"
[23] "green onions, potatoes, chives, tomatoes"
[24] "I enjoy the variety as the items become available. Loved purchasing extra plants for home patio garden. They were excellent quality!" [25] "squash varieties"
[26] "Broccoli, cauliflower, fruits"

Any recommendations to make the Market more accessible?

[1] "Offer more times"
[2] "A more convenient location (especially for individuals with mobility struggles and to promote community outreach)"
[3] "Additional permanent staffing to help them with the at times overwhelming/blessed abundance of a harvest,"
[4] "I am not usually on campus on Fridays so perhaps other days would be better. I'm not sure if you take

credit cards or apple pay but I don't usually have cash. It's also nice to know specifically what is available and maybe the emails say, but I get so many emails that it's hard to read all of them!" [5] "You're doing amazing work! Thank you!!!!!"

[1] "I consistently struggle to make it to the Market during the small time window when it is offered each week. I was more successful during the previous year, when I believe the window was for 2 hours instead of one."
[2] "Perhaps longer time period. The current time period can be during some students classes and they're unable to get to the market in time to browse."
[3] "I was t able to come this fall because of an off campus Friday afternoon work conflict"
[4] "Different times for sales, but I know that's hard"
[5] "Having it in a parking lot instead of in grass would make it easier on rainy/muddy days and for canes/wheelchairs/strollers."
[6] "I would say to move the time maybe around lunch time? I think that later in the day on Friday can sometimes make it difficult to attend."
[7] "Too far away from main buildings. Only open when we should be working. Maybe open during lunch times in a more central location."
[8] "Timing is the biggest issue I have, perhaps more short windows, or a longer window?"
[9] "more hours!"
[10] "I wonder about moving the location to a more central spot, but I also recognize that might be too hard on the team."
[11] "Have a subscription model where individuals sign up for a season's worth of whatever is available that week."
[12] "Have it later in the afternoon"
[13] "Maybe having it at a different time that would be more accessible for students who have late afternoon classes"
[14] "have a golf cart or something running between the market and the farm if people can pick vegetables"
[15] "Have the location closer to the center of campus (or have transportation, like a golf cart, for those who would struggle walking that distance)"
[16] "Maybe longer or more varied hours? (Although that might be too hard.)"
[17] "It might help to not always have it at exactly the same time every week (although I realize that might cause confusion)"
[18] "While the location of the market on campus is more accessible than going all the way to the farm site, it is a bit of a walk on hot, summer days and doesn't easily have a way for people to drive up to it and the ground could be difficult for some to navigate. This is based on the location the last time I went. It is possible that it has moved since then." [19] "People come and access in May ways which suit their abilities."
[20] "If the goal is to extend more outreach to south bend community members, it would help to have the stand (or another stand) closer to the entrance on flat, easily navigated surfaces."